WIKIPEDIA
The Free Encyclopedia

History of artificial intelligence

The **history of artificial intelligence** (**AI**) began in antiquity, with myths, stories and rumors of artificial beings endowed with intelligence or consciousness by master craftsmen. The seeds of modern AI were planted by philosophers who attempted to describe the process of human thinking as the mechanical manipulation of symbols. This work culminated in the invention of the programmable digital computer in the 1940s, a machine based on the abstract essence of mathematical reasoning. This device and the ideas behind it inspired a handful of scientists to begin seriously discussing the possibility of building an electronic brain.

The field of AI research was founded at a workshop held on the campus of Dartmouth College in the U.S. during the summer of 1956.[1] Those who attended would become the leaders of AI research for decades. Many of them predicted that a machine as intelligent as a human being would exist in no more than a generation, and they were given millions of dollars to make this vision come true.[2]

Eventually, it became obvious that researchers had grossly underestimated the difficulty of the project.[3] In 1974, in response to the criticism from James Lighthill and ongoing pressure from the U.S. Congress, the U.S. and British Governments stopped funding undirected research into artificial intelligence. Seven years later, a visionary initiative by the Japanese Government inspired governments and industry to provide AI with billions of dollars, but by the late 1980s the investors became disillusioned and withdrew funding again. The difficult years that followed would later be known as an "AI winter". AI was criticized in the press and avoided by industry until the mid-2000s, but research and funding continued to grow under other names.

In the 1990s and early 2000s machine learning was applied to many problems in academia and industry. The success was due to the availability powerful computer hardware, the collection of immense data sets and the application of solid mathematical methods. In 2012, deep learning proved to be a breakthrough technology, eclipsing all other methods. The transformer architecture debuted in 2017 and was used to produce impressive generative AI applications. Investment in AI boomed in the 2020s.

Precursors

Mythical, fictional, and speculative precursors

Myth and legend

In Greek mythology, Talos was a giant constructed of bronze who acted as guardian for the island of Crete. He would throw boulders at the ships of invaders and would complete 3 circuits around the island's perimeter daily.[4] According to pseudo-Apollodorus' *Bibliotheke*, Hephaestus forged Talos with the aid of a cyclops and presented the automaton as a gift to Minos.[5] In the Argonautica, Jason and the Argonauts defeated him by way of a single plug near his foot which, once removed, allowed the vital ichor to flow out from his body and left him inanimate.[6]

Pygmalion was a legendary king and sculptor of Greek mythology, famously represented in Ovid's *Metamorphoses*. In the 10th book of Ovid's narrative poem, Pygmalion becomes disgusted with women when he witnesses the way in which the Propoetides prostitute themselves.[7] Despite this, he makes offerings at the temple of Venus asking the goddess to bring to him a woman just like a statue he carved.

Medieval legends of artificial beings

Depiction of a homunculus from Goethe's Faust

In *Of the Nature of Things*, written by the Swiss alchemist Paracelsus, he describes a procedure that he claims can fabricate an "artificial man". By placing the "sperm of a man" in horse dung, and feeding it the "Arcanum of Mans blood" after 40 days, the concoction will become a living infant.[8]

The earliest written account regarding golem-making is found in the writings of Eleazar ben Judah of Worms in the early 13th century.[9][10] During the Middle Ages, it was believed that the animation of a Golem could be achieved by insertion of a piece of paper with any of God's names on it, into the mouth of the clay figure.[11] Unlike legendary automata like Brazen Heads,[12] a Golem was unable to speak.[13]

Takwin, the artificial creation of life, was a frequent topic of Ismaili alchemical manuscripts, especially those attributed to Jabir ibn Hayyan. Islamic alchemists attempted to create a broad range of life through their work, ranging from plants to animals.[14]

In Faust: The Second Part of the Tragedy by Johann Wolfgang von Goethe, an alchemically fabricated homunculus, destined to live forever in the flask in which he was made, endeavors to be born into a full human body. Upon the initiation of this transformation, however, the flask shatters and the homunculus dies.[15]

Modern fiction

By the 19th century, ideas about artificial men and thinking machines were developed in fiction, as in Mary Shelley's *Frankenstein* or Karel Čapek's *R.U.R. (Rossum's Universal Robots)*,[16] and speculation, such as Samuel Butler's "Darwin among the Machines",[17] and in real-world instances, including Edgar Allan Poe's "Maelzel's Chess Player".[18] AI is a common topic in science fiction through the present.[19]

Automata

Al-Jazari's programmable automata (1206 CE)

Realistic humanoid automata were built by craftsman from many civilizations, including Yan Shi,[20] Hero of Alexandria,[21] Al-Jazari,[22] Haroun al-Rashid, [23] Jacques de Vaucanson,[24][25] Leonardo Torres y Quevedo,[26] Pierre Jaquet-Droz and Wolfgang von Kempelen.[27][28]

The oldest known automata were the sacred statues of ancient Egypt and Greece.[29][30] The faithful believed that craftsman had imbued these figures with very real minds, capable of wisdom and emotion—Hermes Trismegistus wrote that "by discovering the true nature of the gods, man has been able to reproduce it".[31] English scholar Alexander Neckham asserted that the Ancient Roman poet Virgil had built a palace with automaton statues.[32]

During the early modern period, these legendary automata were said to possess the magical ability to answer questions put to them. The late medieval alchemist and proto-protestant Roger Bacon was purported to have fabricated a brazen head, having developed a legend of having been a wizard.[33][34] These legends were similar to the Norse myth of the Head of Mímir. According to legend, Mímir was known for his intellect and wisdom, and was beheaded in the Æsir-Vanir War. Odin is said to have "embalmed" the head with herbs and spoke incantations over it such that Mímir's head remained able to speak wisdom to Odin. Odin then kept the head near him for counsel.[35]

Formal reasoning

Artificial intelligence is based on the assumption that the process of human thought can be mechanized. The study of mechanical—or "formal"—reasoning has a long history. Chinese, Indian and Greek philosophers all developed structured methods of formal deduction by the first millennium BCE. Their ideas were developed over the centuries by philosophers such as Aristotle (who gave a formal analysis of the syllogism), Euclid (whose *Elements* was a model of formal reasoning), al-Khwārizmī (who developed algebra and gave his name to "algorithm") and European scholastic philosophers such as William of Ockham and Duns Scotus.[36]

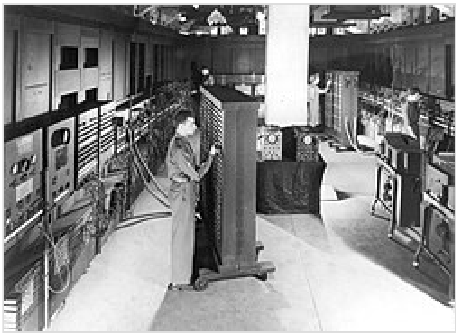
Spanish philosopher Ramon Llull (1232–1315) developed several *logical machines* devoted to the production of knowledge by logical means;[37] Llull described his machines as mechanical entities that could combine basic and undeniable truths by simple logical operations, produced by the machine by mechanical meanings, in such ways as to produce all the possible knowledge.[38] Llull's work had a great influence on Gottfried Leibniz, who redeveloped his ideas.[39]

Gottfried Leibniz, who speculated that human reason could be reduced to mechanical calculation

In the 17th century, Leibniz, Thomas Hobbes and René Descartes explored the possibility that all rational thought could be made as systematic as algebra or geometry.[40] Hobbes famously wrote in *Leviathan*: "reason is nothing but reckoning".[41] Leibniz envisioned a universal language of reasoning, the *characteristica universalis*, which would reduce argumentation to calculation so that "there would be no more need of disputation between two philosophers than between two accountants. For it would suffice to take their pencils in hand, down to their slates, and to say each other (with a friend as witness, if they liked): *Let us calculate*."[42] These philosophers had begun to articulate the physical symbol system hypothesis that would become the guiding faith of AI research.

The study of mathematical logic provided the essential breakthrough that made artificial intelligence seem plausible. The foundations had been set by such works as Boole's *The Laws of Thought* and Frege's *Begriffsschrift*. Building on Frege's system, Russell and Whitehead presented a formal treatment of the foundations of mathematics in their masterpiece, the *Principia Mathematica* in 1913. Inspired by Russell's success, David Hilbert challenged mathematicians of the 1920s and 30s to answer this fundamental question: "can all of mathematical reasoning be formalized?"[36] His question was answered by Gödel's incompleteness proof, Turing's machine and Church's Lambda calculus.[36][a]

Their answer was surprising in two ways. First, they proved that there were, in fact, limits to what mathematical logic could accomplish. But second (and more important for AI) their work suggested that, within these limits, *any* form of mathematical reasoning could be mechanized. The Church-Turing thesis implied that a mechanical device, shuffling symbols as simple as 0 and 1, could imitate any conceivable process of mathematical deduction.[36] The key insight was the Turing machine—a simple theoretical construct that captured the essence of abstract symbol manipulation.[45] This invention would inspire a handful of scientists to begin discussing the possibility of thinking machines.



US Army photo of the ENIAC at the Moore School of Electrical Engineering[44]

Computer science

Calculating machines were designed or built in antiquity and throughout history by many people, including Gottfried Leibniz,[46] Joseph Marie Jacquard,[47] Charles Babbage,[48] Percy Ludgate,[49] Leonardo Torres Quevedo,[50] Vannevar Bush,[51] and others. Ada Lovelace speculated that Babbage's machine was "a thinking or ... reasoning machine", but warned "It is desirable to guard against the possibility of exaggerated ideas that arise as to the powers" of the machine.[52][53]

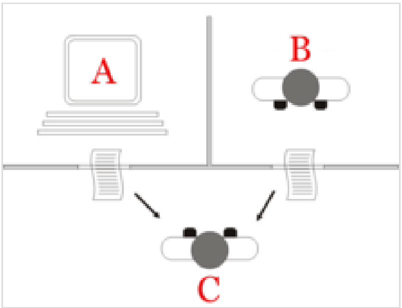
The first modern computers were the massive machines of the Second World War (such as Konrad Zuse's Z3, Alan Turing's Heath Robinson and Colossus, Atanasoff and Berry's and ABC and ENIAC at the University of Pennsylvania).[54] ENIAC was based on the theoretical foundation laid by Alan Turing and developed by John von Neumann,[55] and proved to be the most influential.[54]

Birth of artificial intelligence (1941-56)

The IBM 702: a computer used by the first generation of AI researchers.

The earliest research into thinking machines was inspired by a confluence of ideas that became prevalent in the late 1930s, 1940s, and early 1950s. Recent research in neurology had shown that the brain was an electrical network of neurons that fired in all-or-nothing pulses. Norbert Wiener's cybernetics described control and stability in electrical networks. Claude Shannon's information theory described digital signals (i.e., all-or-nothing signals). Alan Turing's theory of computation showed that any form of computation could be described digitally. The close relationship between these ideas suggested that it might be possible to construct an "electronic brain".

In the 1940s and 50s, a handful of scientists from a variety of fields (mathematics, psychology, engineering, economics and political science) explored several research directions that would be vital to later AI research.[56] Alan Turing was among the first people to seriously investigate the theoretical possibility of "machine intelligence".[57] The field of "artificial intelligence research" was founded as an academic discipline in 1956.[58]



Turing test[59]

Turing Test

In 1950 Turing published a landmark paper "Computing Machinery and Intelligence", in which he speculated about the possibility of creating machines that think.[60][b] In the paper, he noted that "thinking" is difficult to define and devised his famous Turing Test: If a machine could carry on a conversation (over a teleprinter) that was indistinguishable from a conversation with a human being, then it was reasonable to say that the machine was "thinking".[61] This simplified version of the problem allowed Turing to argue convincingly that a "thinking machine" was at least *plausible* and the paper answered all the most common objections to the proposition.[62] The Turing Test was the first serious proposal in the philosophy of artificial intelligence.

Artificial neural networks

Walter Pitts and Warren McCulloch analyzed networks of idealized artificial neurons and showed how they might perform simple logical functions in 1943.[63][64] They were the first to describe what later researchers would call a neural network.[65] The paper was influenced by Turing's paper 'On Computable Numbers' from 1936 using similar two-state boolean 'neurons', but was the first to apply it to neuronal function.[57] One of the students inspired by Pitts and McCulloch was Marvin Minsky who was a 24-year-old graduate student at the time. In 1951 Minsky and Dean Edmonds built the first neural net machine, the SNARC.[66] Minsky would later become one of the most important leaders and innovators in AI.

Cybernetic robots

Experimental robots such as W. Grey Walter's turtles and the Johns Hopkins Beast, were built in the 1950s. These machines did not use computers, digital electronics or symbolic reasoning; they were controlled entirely by analog circuitry.[67]

Game AI

In 1951, using the Ferranti Mark 1 machine of the University of Manchester, Christopher Strachey wrote a checkers program and Dietrich Prinz wrote one for chess.[68] Arthur Samuel's checkers program, the subject of his 1959 paper "Some Studies in Machine Learning Using the Game of Checkers", eventually achieved sufficient skill to challenge a respectable amateur.[69] Game AI would continue to be used as a measure of progress in AI throughout its history.

Symbolic reasoning and the Logic Theorist

When access to digital computers became possible in the mid-fifties, a few scientists instinctively recognized that a machine that could manipulate numbers could also manipulate symbols and that the manipulation of symbols could well be the essence of human thought. This was a new approach to creating thinking machines.[70]

Herbert Simon (left) in a chess match against Allen Newell c. 1958

In 1955, Allen Newell and future Nobel Laureate Herbert A. Simon created the "Logic Theorist", with help from J. C. Shaw. The program would eventually prove 38 of the first 52 theorems in Russell and Whitehead's *Principia Mathematica*, and find new and more elegant proofs for some.[71] Simon said that they had "solved the venerable mind/body problem, explaining how a system composed of matter can have the properties of mind."[72] This was an early statement of the philosophical position John Searle would later call "Strong AI": that machines can contain minds just as human bodies do.[73] The symbolic reasoning paradigm they introduced would dominate AI research and funding until the middle 90s, as well as inspire the cognitive revolution.

Dartmouth Workshop

The Dartmouth workshop of 1956 was a pivotal event that marked the formal inception of AI as an academic discipline.[74] It was organized by Marvin Minsky, John McCarthy, with the support of two senior scientists Claude Shannon and Nathan Rochester of IBM. The proposal for the conference stated they intended to test the assertion that "every aspect of learning or any other feature of intelligence can be so precisely described that a machine can be made to simulate it".[75] The term "Artificial Intelligence" was introduced by John McCarthy at the workshop.[76][c] The participants included Ray Solomonoff, Oliver Selfridge, Trenchard More, Arthur Samuel, Allen Newell and Herbert A. Simon, all of whom would create important programs during the first decades of AI research.[78] At the workshop Newell and Simon debuted the "Logic Theorist".[79] The workshop was the moment that AI gained its name, its mission, its first major success and its key players, and is widely considered the birth of AI.[80]

Cognitive revolution

In the fall of 1956, Newell and Simon also presented the Logic Theorist at a meeting of the Special Interest Group in Information Theory at the Massachusetts Institute of Technology (MIT). At the same meeting, Noam Chomsky discussed his generative grammar, and George Miller described his landmark paper "The Magical Number Seven, Plus or Minus Two". Miller wrote "I left the symposium with a conviction, more intuitive than rational, that experimental psychology, theoretical linguistics, and the computer simulation of cognitive processes were all pieces from a larger whole."[81]

This meeting was the beginning of the "cognitive revolution"—an interdisciplinary paradigm shift in psychology, philosophy, computer science and neuroscience. It inspired the creation of the sub-fields of symbolic artificial intelligence, generative linguistics, cognitive science, cognitive psychology, cognitive neuroscience and the philosophical schools of computationalism and functionalism. All these fields used related tools to model the mind and results discovered in one field were relevant to the others.

The cognitive approach allowed researchers to consider "mental objects" like thoughts, plans, goals, facts or memories, often analyzed using high level symbols in functional networks. These objects had been forbidden as "unobservable" by earlier paradigms such as behaviorism. Symbolic mental objects would become the major focus of AI research and funding for the next several decades.

Early successes (1956-1974)

The programs developed in the years after the Dartmouth Workshop were, to most people, simply "astonishing":[82] computers were solving algebra word problems, proving theorems in geometry and learning to speak English. Few at the time would have believed that such "intelligent" behavior by machines was possible at all.[83] Researchers expressed an intense optimism in private and in print, predicting that a fully intelligent machine would be built in less than 20 years.[84] Government agencies like the Defense Advanced Research Projects Agency (DARPA, then known as "ARPA") poured money into the field.[85] Artificial Intelligence laboratories were set up at a number of British and US universities in the latter 1950s and early 1960s.[57]

Approaches

There were many successful programs and new directions in the late 50s and 1960s. Among the most influential were these:

Reasoning as search

Many early AI programs used the same basic algorithm. To achieve some goal (like winning a game or proving a theorem), they proceeded step by step towards it (by making a move or a deduction) as if searching through a maze, backtracking whenever they reached a dead end.

The principal difficulty was that, for many problems, the number of possible paths through the "maze" was astronomical (a situation known as a "combinatorial explosion"). Researchers would reduce the search space by using heuristics that would eliminate paths that were unlikely to lead to a solution.[86]

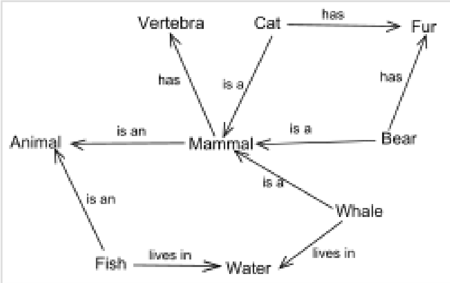
Newell and Simon tried to capture a general version of this algorithm in a program called the "General Problem Solver".[87] Other "searching" programs were able to accomplish impressive tasks like solving problems in geometry and algebra, such as Herbert Gelernter's Geometry Theorem Prover (1958) and Symbolic Automatic Integrator (SAINT), written by Minsky's student James Slagle in 1961.[88] Other programs searched through goals and subgoals to plan actions, like the STRIPS system developed at Stanford to control the behavior of the robot Shakey.[89]

Natural language

An important goal of AI research is to allow computers to communicate in natural languages like English. An early success was Daniel Bobrow's program STUDENT, which could solve high school algebra word problems.[90]

A semantic net represents concepts (e.g. "house", "door") as nodes, and relations among concepts as links between the nodes (e.g. "has-a"). The first AI program to use a semantic net was written by Ross Quillian[91] and the most successful (and controversial) version was Roger Schank's Conceptual dependency theory.[92]

Joseph Weizenbaum's ELIZA could carry out conversations that were so realistic that users occasionally were fooled into thinking they were communicating with a human being and not a computer program (see ELIZA effect). But in fact, ELIZA simply gave a canned response or repeated back what was said to it, rephrasing its response with a few grammar rules. ELIZA was the first chatbot.[93]



An example of a semantic network

Micro-worlds

In the late 60s, Marvin Minsky and Seymour Papert of the MIT AI Laboratory proposed that AI research should focus on artificially simple situations known as micro-worlds. They pointed out that in successful sciences like physics, basic principles were often best understood using simplified models like frictionless planes or perfectly rigid bodies. Much of the research focused on a "blocks world," which consists of colored blocks of various shapes and sizes arrayed on a flat surface.[94]

This paradigm led to innovative work in machine vision by Gerald Sussman, Adolfo Guzman, David Waltz (who invented "constraint propagation"), and especially Patrick Winston. At the same time, Minsky and Papert built a robot arm that could stack blocks, bringing the blocks world to life. Terry Winograd's SHRDLU could communicate in ordinary English sentences about the micro-world, plan operations and execute them.[95]

Perceptrons

In the 1960s the cognitive revolution was highly influential and most funding was directed towards laboratories researching symbolic AI. However, there was one exception: the perceptron, a single-layer neural network introduced in 1958 by Frank Rosenblatt (who had been a schoolmate of Marvin Minsky at the Bronx High School of Science). Like most AI researchers, he was optimistic about their power, predicting that a perceptron “may eventually be able to learn, make decisions, and translate languages."[98]

The Mark 1 Perceptron.[96][97]

Rosenblatt was primarily funded by Office of Naval Research.[99] Bernard Widrow and his student Ted Hoff built ADALINE (1960) and MADALINE (1962), which had up to 1000 adjustable weights.[100] A group at Stanford Research Institute led by Charles A. Rosen and Alfred E. (Ted) Brain built two neural network machines named MINOS I (1960) and II (1963), mainly funded by U.S. Army Signal Corps. MINOS II[101] had 6600 adjustable weights,[102] and was controlled with an SDS 910 computer in a configuration named MINOS III (1968), which could classify symbols on army maps, and recognize hand-printed characters on Fortran coding sheets.[103][104][105]

Most of neural network research during this early period involved building and using bespoke hardware, rather than simulation on digital computers. The hardware diversity was particularly clear in the different technologies used in implementing the adjustable weights. The perceptron machines and the SNARC used potentiometers moved by electric motors. ADALINE used memistors adjusted by electroplating, though they also used simulations on an IBM 1620 computer. The MINOS machines used ferrite cores with multiple holes in them that could be individually blocked, with the degree of blockage representing the weights.[106]

However, partly due to lack of results and partly due to competition from symbolic AI research, the MINOS project ran out of funding in 1966. Rosenblatt failed to secure continued funding in the 1960s.[106] In 1969, research came to a sudden halt with the publication of Minsky and Papert's 1969 book *Perceptrons*.[107] It suggested that there were severe limitations to what perceptrons could do and that Rosenblatt's predictions had been grossly exaggerated. The effect of the book was that virtually no research was funded in connectionism for 10 years.[98] The competition for government funding ended with the victory of symbolic AI approaches over neural networks.[105][106]

Minsky (who had worked on SNARC) became a staunch objector to pure connectionist AI. Widrow (who had worked on ADALINE) turned to adaptive signal processing. The SRI group (which worked on MINOS) turned to symbolic AI and robotics.[105][106]

The main problem was the inability to train multilayered networks (versions of backpropagation had already been used in other fields but it was unknown to these researchers). Rosenblatt attempted to gather funds for building larger perceptron machines, but he died in a boating accident in 1971.[98]

Optimism

The first generation of AI researchers made these predictions about their work:

- 1958, H. A. Simon and Allen Newell: "within ten years a digital computer will be the world's chess champion" and "within ten years a digital computer will discover and prove an important new mathematical theorem."[108]
- 1965, H. A. Simon: "machines will be capable, within twenty years, of doing any work a man can do."[109]
- 1967, Marvin Minsky: "Within a generation ... the problem of creating 'artificial intelligence' will substantially be solved."[110]
- 1970, Marvin Minsky (in *Life* Magazine): "In from three to eight years we will have a machine with the general intelligence of an average human being."[111]

Financing

In June 1963, MIT received a $2.2 million grant from the newly created Advanced Research Projects Agency (ARPA, later known as DARPA). The money was used to fund project MAC which subsumed the "AI Group" founded by Minsky and McCarthy five years earlier. DARPA continued to provide $3 million each year until the 70s.[112] DARPA made similar grants to Newell and Simon's program at Carnegie Mellon University and to Stanford University's AI Lab, founded by John McCarthy in 1963.[113] Another important AI laboratory was established at Edinburgh University by Donald Michie in 1965.[114] These four institutions would continue to be the main centers of AI research and funding in academia for many years.[115]

The money was given with few strings attached: J. C. R. Licklider, then the director of ARPA, believed that his organization should "fund people, not projects!" and allowed researchers to pursue whatever directions might interest them.[116] This created a freewheeling atmosphere at MIT that gave birth to the hacker culture,[117] but this "hands off" approach did not last.

First AI Winter (1974–1980)

In the 1970s, AI was subject to critiques and financial setbacks. AI researchers had failed to appreciate the difficulty of the problems they faced. Their tremendous optimism had raised public expectations impossibly high, and when the promised results failed to materialize, funding targeted at AI was severely reduced.[118] The lack of success indicated the techniques being used by AI researchers at the time were insufficient to achieve their goals.[119][120]

These setbacks did not affect the growth and progress of the field, however. The funding cuts only impacted a handful of major laboratories[121] and the critiques were largely ignored.[122] General public interest in the field continued to grow,[121] the number of researchers increased dramatically,[121] and new ideas were explored in logic programming, commonsense reasoning and many other areas. Historian Thomas Haigh argues that there was no winter,[121] and AI researcher Nils Nilsson described this period as the most "exciting" time to work in AI.[123]

Problems

In the early seventies, the capabilities of AI programs were limited. Even the most impressive could only handle trivial versions of the problems they were supposed to solve; all the programs were, in some sense, "toys".[124] AI researchers had begun to run into several limits that would be only conquered decades later, and others that still stymie the field in the 2020s:[125]

- **Limited computer power.** There was not enough memory or processing speed to accomplish anything truly useful.[126] For example: Ross Quillian's successful work on natural language was demonstrated with a vocabulary of only 20 words, because that was all that would fit in memory.[127] Hans Moravec argued in 1976 that computers were still millions of times too weak to exhibit intelligence. He suggested an analogy: artificial intelligence requires computer power in the same way that aircraft require horsepower. Below a certain threshold, it's impossible, but, as power increases, eventually it could become easy. "With enough horsepower," he wrote, "anything will fly".[128][d]
- **Intractability** and the **combinatorial explosion**. In 1972 Richard Karp (building on Stephen Cook's 1971 theorem) showed there are many problems that can only be solved in exponential time. Finding optimal solutions to these problems requires extraordinary amounts of computer time, except when the problems are trivial. This meant that many of the "toy" solutions used by AI would never scale to useful systems.[130]
- **Commonsense knowledge** and **reasoning**. Many important artificial intelligence applications like vision or natural language require enormous amounts of information about the world: the program needs to have some idea of what it might be looking at or what it is talking about. This requires that the program know most of the same things about the world that a child does. Researchers soon discovered that this was a vast amount of information. No one in 1970 could build a database large enough and no one knew how a program might learn so much information.[131]
- **Moravec's paradox**: Proving theorems and solving geometry problems is comparatively easy for computers, but a supposedly simple task like recognizing a face or crossing a room without bumping into anything is extremely difficult, and research into vision and robotics made little progress in the early 1970s.[132]
- The **frame** and **qualification problems**. AI researchers (like John McCarthy) who used logic discovered that they could not represent ordinary deductions that involved planning or default reasoning without making changes to the structure of logic itself. They developed new logics (like non-monotonic logics and modal logics) to try to solve the problems.[133]

Decrease in funding

The agencies which funded AI research, such as the British government, DARPA and the National Research Council (NRC) became frustrated with the lack of progress and eventually cut off almost all funding for undirected AI research. The pattern began in 1966 when the Automatic Language Processing Advisory Committee (ALPAC) report criticized machine translation efforts. After spending $20 million, the NRC ended all support.[134] In 1973, the Lighthill report on the state of AI research in the UK criticized the failure of AI to achieve its "grandiose objectives" and led to the dismantling of AI research in that country.[135] (The report specifically mentioned the combinatorial explosion problem as a reason for AI's failings.)[136][e] DARPA was deeply disappointed with researchers working on the Speech Understanding Research program at CMU and canceled an annual grant of $3 million.[137][f]

Hans Moravec blamed the crisis on the unrealistic predictions of his colleagues. "Many researchers were caught up in a web of increasing exaggeration." [138][g] However, there was another issue: since the passage of the Mansfield Amendment in 1969, DARPA had been under increasing pressure to fund "mission-oriented direct research, rather than basic undirected research". Funding for the creative, freewheeling exploration that had gone on in the 60s would not come from DARPA, which instead directed money at specific projects with clear objectives, such as autonomous tanks and battle management systems. [139][h]

The major laboratories (MIT, Stanford and CMU) had been receiving generous support from the U.S. military, and when it was withdrawn, these were the only places that were seriously impacted by the budget cuts. The thousands of researchers outside these institutions and the many more thousands that were joining the field were unaffected.[121]

Philosophical and ethical critiques

Several philosophers had strong objections to the claims being made by AI researchers. One of the earliest was John Lucas, who argued that Gödel's incompleteness theorem showed that a formal system (such as a computer program) could never see the truth of certain statements, while a human being could.[140] Hubert Dreyfus ridiculed the broken promises of the 1960s and critiqued the assumptions of AI, arguing that human reasoning actually involved very little "symbol processing" and a great deal of embodied, instinctive, unconscious "know how".[141][142] John Searle's Chinese Room argument, presented in 1980, attempted to show that a program could not be said to "understand" the symbols that it uses (a quality called "intentionality"). If the symbols have no meaning for the machine, Searle argued, then the machine can not be described as "thinking".[143]

These critiques were not taken seriously by AI researchers. Problems like intractability and commonsense knowledge seemed much more immediate and serious. It was unclear what difference "know how" or "intentionality" made to an actual computer program. MIT's Minsky said of Dreyfus and Searle "they misunderstand, and should be ignored."[144] Dreyfus, who also taught at MIT, was given a cold shoulder: he later said that AI researchers "dared not be seen having lunch with me."[145] Joseph Weizenbaum, the author of ELIZA, was also an outspoken critic of Dreyfus' positions, but he "deliberately made it plain that [his AI colleagues' treatment of Dreyfus] was not the way to treat a human being,"[146] and was unprofessional and childish.[147]

Weizenbaum began to have serious ethical doubts about AI when Kenneth Colby wrote a "computer program which can conduct psychotherapeutic dialogue" based on ELIZA.[148] Weizenbaum was disturbed that Colby saw a mindless program as a serious therapeutic tool. A feud began, and the situation was not helped when Colby did not credit Weizenbaum for his contribution to the program. In 1976, Weizenbaum published *Computer Power and Human Reason* which argued that the misuse of artificial intelligence has the potential to devalue human life.[149]

Logic at Stanford, CMU and Edinburgh

Logic was introduced into AI research as early as 1959, by John McCarthy in his Advice Taker proposal.[150] In 1963, J. Alan Robinson had discovered a simple method to implement deduction on computers, the resolution and unification algorithm. However, straightforward implementations, like those attempted by McCarthy and his students in the late 1960s, were especially intractable: the programs required astronomical numbers of steps to prove simple theorems.[151] A more fruitful approach to logic was developed in the 1970s by Robert Kowalski at the University of Edinburgh, and soon this led to the collaboration with French researchers Alain Colmerauer and Philippe Roussel who created the successful logic programming language Prolog.[152] Prolog uses a subset of logic (Horn clauses, closely related to "rules" and "production rules") that permit tractable computation. Rules would continue to be influential, providing a foundation for Edward Feigenbaum's expert systems and the continuing work by Allen Newell and Herbert A. Simon that would lead to Soar and their unified theories of cognition.[153]

Critics of the logical approach noted, as Dreyfus had, that human beings rarely used logic when they solved problems. Experiments by psychologists like Peter Wason, Eleanor Rosch, Amos Tversky, Daniel Kahneman and others provided proof.[154] McCarthy responded that what people do is irrelevant. He argued that what is really needed are machines that can solve problems—not machines that think as people do.[155]

MIT's "anti-logic" approach

Among the critics of McCarthy's approach were his colleagues across the country at MIT. Marvin Minsky, Seymour Papert and Roger Schank were trying to solve problems like "story understanding" and "object recognition" that *required* a machine to think like a person. In order to use ordinary concepts like "chair" or "restaurant" they had to make all the same illogical assumptions that people normally made. Unfortunately, imprecise concepts like these are hard to represent in logic. Gerald Sussman observed that "using precise language to describe essentially imprecise concepts doesn't make them any more precise."[156] Schank described their "anti-logic" approaches as "scruffy", as opposed to the "neat" paradigms used by McCarthy, Kowalski, Feigenbaum, Newell and Simon.[157]

In 1975, in a seminal paper, Minsky noted that many of his fellow researchers were using the same kind of tool: a framework that captures all our common sense assumptions about something. For example, if we use the concept of a bird, there is a constellation of facts that immediately come to mind: we might assume that it flies, eats worms and so on. We know these facts are not always true and that deductions using these facts will not be "logical", but these structured sets of assumptions are part of the *context* of everything we say and think. He called these structures "frames". Schank used a version of frames he called "scripts" to successfully answer questions about short stories in English.[158]

The emergence of non-monotonic logics

The logicians rose to the challenge. Pat Hayes claimed that "most of 'frames' is just a new syntax for parts of first-order logic." But he noted that "there are one or two apparently minor details which give a lot of trouble, however, especially defaults".[159] In the meanwhile, Ray Reiter admitted that "conventional logics, such as first-order logic, lack the expressive power to adequately represent the knowledge required for reasoning by default".[160] He proposed augmenting first-order logic with a closed world assumption that a conclusion holds (by default) if its contrary cannot be shown. He showed how such an assumption corresponds to the common sense assumption made in reasoning with frames. He also showed that it has its "procedural equivalent" as negation as failure in Prolog.

The closed world assumption, as formulated by Reiter, "is not a first-order notion. (It is a meta notion.)"[160] However, Keith Clark showed that negation as *finite failure* can be understood as reasoning implicitly with definitions in first-order logic including a unique name assumption that different terms denote different individuals.[161]

During the late 1970s and throughout the 1980s, a variety of logics and extensions of first-order logic were developed both for negation as failure in logic programming and for default reasoning more generally. Collectively, these logics have become known as non-monotonic logics.

Boom (1980–1987)

In the 1980s, a form of AI program called "expert systems" was adopted by corporations around the world and knowledge became the focus of mainstream AI research. Governments provided substantial funding, such as Japan's fifth generation computer project and the U.S. Strategic Computing Initiative.

Although symbolic knowledge representation and logical reasoning produced these useful applications in the 80s, it was still unable to solve problems in perception, robotics, learning and common sense. A small number of scientists and engineers began to doubt that the symbolic approach would ever be sufficient for these tasks and developed other approaches, such as *connectionism*, robotics, and *soft* computing.

Expert systems become widely used

An expert system is a program that answers questions or solves problems about a specific domain of knowledge, using logical rules that are derived from the knowledge of experts. The earliest examples were developed by Edward Feigenbaum and his students. Dendral, begun in 1965, identified compounds from spectrometer readings. MYCIN, developed in 1972, diagnosed infectious blood diseases. They demonstrated the feasibility of the approach.[162]

Expert systems restricted themselves to a small domain of specific knowledge (thus avoiding the commonsense knowledge problem) and their simple design made it relatively easy for programs to be built and then modified once they were in place. All in all, the programs proved to be *useful*: something that AI had not been able to achieve up to this point.[163]

In 1980, an expert system called XCON was completed at CMU for the Digital Equipment Corporation. It was an enormous success: it was saving the company 40 million dollars annually by 1986.[164] Corporations around the world began to develop and deploy expert systems and by 1985 they were spending over a billion dollars on AI, most of it to in-house AI departments.[165] An industry grew up to support them, including hardware companies like Symbolics and Lisp Machines and software companies such as IntelliCorp and Aion.[166]

Government funding increases

In 1981, the Japanese Ministry of International Trade and Industry set aside $850 million for the Fifth generation computer project. Their objectives were to write programs and build machines that could carry on conversations, translate languages, interpret pictures, and reason like human beings.[167] Much to the chagrin of scruffies, they chose Prolog as the primary computer language for the project.[168]

Other countries responded with new programs of their own. The UK began the £350 million Alvey project. A consortium of American companies formed the Microelectronics and Computer Technology Corporation (or "MCC") to fund large scale projects in AI and information technology.[169][170] DARPA responded as well, founding the Strategic Computing Initiative and tripling its investment in AI between 1984 and 1988.[171]

Knowledge revolution

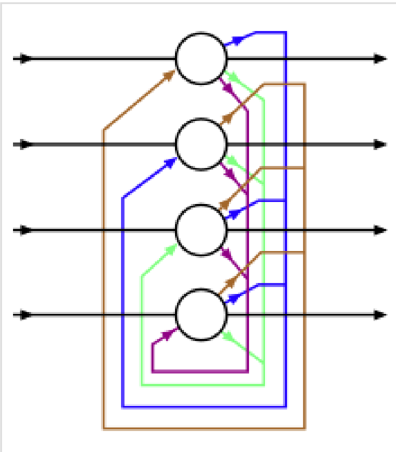
The power of expert systems came from the expert knowledge they contained. They were part of a new direction in AI research that had been gaining ground throughout the 70s. "AI researchers were beginning to suspect—reluctantly, for it violated the scientific canon of parsimony—that intelligence might very well be based on the ability to use large amounts of diverse knowledge in different ways,"[172] writes Pamela McCorduck. "[T]he great lesson from the 1970s was that intelligent behavior depended very much on dealing with knowledge, sometimes quite detailed knowledge, of a domain where a given task lay".[173] Knowledge based systems and knowledge engineering became a major focus of AI research in the 1980s.[174]

The 1980s also saw the birth of Cyc, the first attempt to attack the commonsense knowledge problem directly, by creating a massive database that would contain all the mundane facts that the average person knows. Douglas Lenat, who started and led the project, argued that there is no shortcut — the only way for machines to know the meaning of human concepts is to teach them, one concept at a time, by hand. The project was not expected to be completed for many decades.[175]

Chess playing programs HiTech and Deep Thought defeated chess masters in 1989. Both were developed by Carnegie Mellon University; Deep Thought development paved the way for Deep Blue.[176]

Revival of neural networks: *connectionism*

In 1982, physicist John Hopfield was able to prove that a form of neural network (now called a "Hopfield net") could learn and process information, and provably converges after enough time under any fixed condition. It was a breakthrough, as it was previously thought that nonlinear networks would, in general, evolve chaotically.[177] Around the same time, Geoffrey Hinton and David Rumelhart popularized a method for training neural networks called "backpropagation".[i] These two discoveries helped to revive the exploration of artificial neural networks.[170][178]



A Hopfield net with four nodes

Neural networks, along with several other similar models, received widespread attention after the 1986 publication of the *Parallel Distributed Processing*, a two volume collection of papers edited by Rumelhart and psychologist James McClelland. The new field was christened *connectionism* and there was a considerable debate between advocates of symbolic AI the "connectionists".

In 1990, Yann LeCun at Bell Labs used convolutional neural networks to recognize handwritten digits. The system was used widely in 90s, reading zip codes and personal checks. This was the first genuinely useful application of neural networks.[179][180]

Robotics and embodied reason

Rodney Brooks, Hans Moravec and others argued that, in order to show real intelligence, a machine needs to have a *body* — it needs to perceive, move, survive and deal with the world.[181] Sensorimotor skills are essential to higher level skills such as commonsense reasoning. They can't be efficiently implemented using abstract symbolic reasoning, so AI should solve the problems of perception, mobility, manipulation and survival without using symbolic representation at all. These robotics researchers advocated building intelligence "from the bottom up".[j]

A precursor to this idea was David Marr, who had come to MIT in the late 1970s from a successful background in theoretical neuroscience to lead the group studying vision. He rejected all symbolic approaches (*both* McCarthy's logic and Minsky's frames), arguing that AI needed to understand the physical machinery of vision from the bottom up before any symbolic processing took place. (Marr's work would be cut short by leukemia in 1980.)[183]

In his 1990 paper "Elephants Don't Play Chess,"[184] robotics researcher Brooks took direct aim at the physical symbol system hypothesis, arguing that symbols are not always necessary since "the world is its own best model. It is always exactly up to date. It always has every detail there is to be known. The trick is to sense it appropriately and often enough."[185]

In the 1980s and 1990s, many cognitive scientists also rejected the symbol processing model of the mind and argued that the body was essential for reasoning, a theory called the *embodied mind thesis*.[186]

Bust: second AI winter

The business community's fascination with AI rose and fell in the 1980s in the classic pattern of an economic bubble. As dozens of companies failed, the perception in the business world was that the technology was not viable.[187] The damage to AI's reputation would last into the 21st century. Inside the field there was little agreement on the reasons for AI's failure to fulfill the dream of human level intelligence that had captured the imagination of the world in the 1960s. Together, all these factors helped to fragment AI into competing subfields focused on particular problems or approaches, sometimes even under new names that disguised the tarnished pedigree of "artificial intelligence".[188]

Over the next 20 years, AI consistently delivered working solutions to specific isolated problems. By the late 1990s, it was being used throughout the technology industry, although somewhat behind the scenes. The success was due to increasing computer power, by collaboration with other fields (such as mathematical optimization and statistics) and using the highest standards of scientific accountability. By 2000, AI had achieved some of its oldest goals. The field was both more cautious and more successful than it had ever been.

AI winter

The term "AI winter" was coined by researchers who had survived the funding cuts of 1974 when they became concerned that enthusiasm for expert systems had spiraled out of control and that disappointment would certainly follow.[189] Their fears were well founded: in the late 1980s and early 1990s, AI suffered a series of financial setbacks.

The first indication of a change in weather was the sudden collapse of the market for specialized AI hardware in 1987. Desktop computers from Apple and IBM had been steadily gaining speed and power and in 1987 they became more powerful than the more expensive Lisp machines made by Symbolics and others. There was no longer a good reason to buy them. An entire industry worth half a billion dollars was demolished overnight.[190]

Eventually the earliest successful expert systems, such as XCON, proved too expensive to maintain. They were difficult to update, they could not learn, they were "brittle" (i.e., they could make grotesque mistakes when given unusual inputs), and they fell prey to problems (such as the qualification problem) that had been identified years earlier. Expert systems proved useful, but only in a few special contexts.[191]

In the late 1980s, the Strategic Computing Initiative cut funding to AI "deeply and brutally". New leadership at DARPA had decided that AI was not "the next wave" and directed funds towards projects that seemed more likely to produce immediate results.[192]

By 1991, the impressive list of goals penned in 1981 for Japan's Fifth Generation Project had not been met. Indeed, some of them, like "carry on a casual conversation" had not been met by 2010.[193] As with other AI projects, expectations had run much higher than what was actually possible.[193][194]

Over 300 AI companies had shut down, gone bankrupt, or been acquired by the end of 1993, effectively ending the first commercial wave of AI.[195] In 1994, HP Newquist stated in *The Brain Makers* that "The immediate future of artificial intelligence—in its commercial form—seems to rest in part on the continued success of neural networks."[195]

AI behind the scenes

In the 1990s, algorithms originally developed by AI researchers began to appear as parts of larger systems. AI had solved a lot of very difficult problems[196] and their solutions proved to be useful throughout the technology industry,[197] such as data mining, industrial robotics, logistics,[198] speech recognition,[199] banking software,[200] medical diagnosis[200] and Google's search engine.[201]

The field of AI received little or no credit for these successes in the 1990s and early 2000s. Many of AI's greatest innovations have been reduced to the status of just another item in the tool chest of computer science.[202] Nick Bostrom explains "A lot of cutting edge AI has filtered into general applications, often without being called AI because once something becomes useful enough and common enough it's not labeled AI anymore."[203]

Many researchers in AI in the 1990s deliberately called their work by other names, such as informatics, knowledge-based systems, "cognitive systems" or computational intelligence. In part, this may have been because they considered their field to be fundamentally different from AI, but also the new names help to procure funding. In the commercial world at least, the failed promises of the AI Winter continued to haunt AI research into the 2000s, as the *New York Times* reported in 2005: "Computer scientists and software engineers avoided the term artificial intelligence for fear of being viewed as wild-eyed dreamers."[204][205][206][207]

Milestones and Moore's law

On May 11, 1997, Deep Blue became the first computer chess-playing system to beat a reigning world chess champion, Garry Kasparov.[208] The super computer was a specialized version of a framework produced by IBM, and was capable of processing twice as many moves per second as it had during the first match (which Deep Blue had lost), reportedly 200,000,000 moves per second.[209]

In 2005, a Stanford robot won the DARPA Grand Challenge by driving autonomously for 131 miles along an unrehearsed desert trail.[210] Two years later, a team from CMU won the DARPA Urban Challenge by autonomously navigating 55 miles in an urban environment while responding to traffic hazards and adhering to traffic laws.[211] In February 2011, in a Jeopardy! quiz show exhibition match, IBM's question answering system, Watson, defeated the two best Jeopardy! champions, Brad Rutter and Ken Jennings, by a significant margin.[212]

These successes were not due to some revolutionary new paradigm, but mostly on the tedious application of engineering skill and on the tremendous increase in the speed and capacity of computer by the 90s.[213] In fact, Deep Blue's computer was 10 million times faster than the Ferranti Mark 1 that Christopher Strachey taught to play chess in 1951.[214] This dramatic increase is measured by Moore's law, which predicts that the speed and memory capacity of computers doubles every two years, as a result of metal–oxide–semiconductor (MOS) transistor counts doubling every two years. The fundamental problem of "raw computer power" was slowly being overcome.

Intelligent agents

A new paradigm called "intelligent agents" became widely accepted during the 1990s.[215] Although earlier researchers had proposed modular "divide and conquer" approaches to AI,[216] the intelligent agent did not reach its modern form until Judea Pearl, Allen Newell, Leslie P. Kaelbling, and others brought concepts from decision theory and economics into the study of AI.[217] When the economist's definition of a rational agent was married to computer science's definition of an object or module, the intelligent agent paradigm was complete.

An intelligent agent is a system that perceives its environment and takes actions which maximize its chances of success. By this definition, simple programs that solve specific problems are "intelligent agents", as are human beings and organizations of human beings, such as firms. The intelligent agent paradigm defines AI research as "the study of intelligent agents". This is a generalization of some earlier definitions of AI: it goes beyond studying human intelligence; it studies all kinds of intelligence.[218]

The paradigm gave researchers license to study isolated problems and find solutions that were both verifiable and useful. It provided a common language to describe problems and share their solutions with each other, and with other fields that also used concepts of abstract agents, like economics and control theory. It was hoped that a complete agent architecture (like Newell's SOAR) would one day allow researchers to build more versatile and intelligent systems out of interacting intelligent agents.[217][219]

Probabilistic reasoning and greater rigor

AI researchers began to develop and use sophisticated mathematical tools more than they ever had in the past.[220] There was a widespread realization that many of the problems that AI needed to solve were already being worked on by researchers in fields like mathematics, electrical engineering, economics or operations research. The shared mathematical language allowed both a higher level of collaboration with more established and successful fields and the achievement of results which were measurable and provable; AI had become a more rigorous "scientific" discipline.

Judea Pearl's influential 1988 book[221] brought probability and decision theory into AI. Among the many new tools in use were Bayesian networks, hidden Markov models, information theory, stochastic modeling and classical optimization. Precise mathematical descriptions were also developed for "computational intelligence" paradigms like neural networks and evolutionary algorithms.[222]

Deep learning, big data (2011–2020)

In the first decades of the 21st century, access to large amounts of data (known as "big data"), cheaper and faster computers and advanced machine learning techniques were successfully applied to many problems throughout the economy. In fact, McKinsey Global Institute estimated in their famous paper "Big data: The next frontier for innovation, competition, and productivity" that "by 2009, nearly all sectors in the US economy had at least an average of 200 terabytes of stored data".

By 2016, the market for AI-related products, hardware, and software reached more than $8 billion, and the New York Times reported that interest in AI had reached a "frenzy".[223] The applications of big data began to reach into other fields as well, such as training models in ecology[224] and for various applications in economics.[225] Advances in deep learning (particularly deep convolutional neural networks and recurrent neural networks) drove progress and research in image and video processing, text analysis, and even speech recognition.[226]

The first global AI Safety Summit was held in Bletchley Park, UK, in November 2023 to discuss the short- and long-term risks of AI and the possibility of mandatory and voluntary regulatory frameworks.[227] Twenty-eight countries including the United States, China, and the European Union issued a declaration at the start of the summit, calling for international co-operation to manage the challenges and risks of artificial intelligence.[228][229]

Deep learning

Deep learning is a branch of machine learning that models high level abstractions in data by using a deep graph with many processing layers.[226] According to the Universal approximation theorem, deep-ness isn't necessary for a neural network to be able to approximate arbitrary continuous functions. Even so, there are many problems that are common to shallow networks (such as overfitting) that deep networks help avoid.[230] As such, deep neural networks are able to realistically generate much more complex models as compared to their shallow counterparts.

However, deep learning has problems of its own. A common problem for recurrent neural networks is the vanishing gradient problem, which is where gradients passed between layers gradually shrink and literally disappear as they are rounded off to zero. There have been many methods developed to approach this problem, such as Long short-term memory units.

State-of-the-art deep neural network architectures can sometimes even rival human accuracy in fields like computer vision, specifically on things like the Modified National Institute of Standards and Technology (MNIST) database, and traffic sign recognition.[231]

Language processing engines powered by smart search engines can easily beat humans at answering general trivia questions (such as IBM Watson), and recent developments in deep learning have produced surprising results in competing with humans, specifically in games like Go, and *Doom* (which, being a first-person shooter game, has sparked some controversy).[232][233][234][235]

Big data

Big data refers to a collection of data that cannot be captured, managed, and processed by conventional software tools within a certain time frame. It is a massive amount of decision-making, insight, and process optimization capabilities that require new processing models. In the book "Big Data Era" by Victor Meyer Schonberg and Kenneth Cooke, big data means that instead of random analysis (sample survey), all data is used for analysis. The "5V" characteristics of big data, as proposed by IBM, are: *Volume*, *Velocity*, *Variety*,[236] *Value*,[237] *Veracity*.[238]

The strategic significance of big data technology is not to master huge data information, but to specialize in meaningful data. In other words, if big data is likened to an industry, the key to realizing profitability in this industry is to increase the "process capability" of the data and realize the "value added" of the data through "processing".

Large language models, AI boom (2020–present)

The AI boom started with the initial development of key architectures and algorithms such as the transformer architecture in 2017, leading to the scaling and development of large language models exhibiting human-like traits of reasoning, cognition, attention and creativity. The new AI era has been said to have begun around 2022–2023, with the public release of scaled large language models (LLMs) such as ChatGPT.[239][240][241][242][243]

Large language models

In 2017, the transformer architecture was proposed by Google researchers. It exploits an attention mechanism and later became widely used in large language models.[244]

Foundation models, which are large language models trained on vast quantities of unlabeled data that can be adapted to a wide range of downstream tasks, began to be developed in 2018.

Models such as GPT-3 released by OpenAI in 2020, and Gato released by DeepMind in 2022, have been described as important achievements of machine learning.

In 2023, Microsoft Research tested the GPT-4 large language model with a large variety of tasks, and concluded that "it could reasonably be viewed as an early (yet still incomplete) version of an artificial general intelligence (AGI) system".[245]

See also

- History of artificial neural networks
- History of knowledge representation and reasoning
- History of natural language processing
- Outline of artificial intelligence
- Progress in artificial intelligence
- Timeline of artificial intelligence
- Timeline of machine learning

Notes

a. The Lambda calculus was especially important to AI, since it was an inspiration for Lisp (the most important programming language used in 20th century AI).[43]
b. Alan Turing was thinking about machine intelligence at least as early as 1941, when he circulated a paper on machine intelligence which could be the earliest paper in the field of AI — although it is now lost. His 1950 paper was followed by three radio broadcasts on AI by Turing, the two lectures 'Intelligent Machinery, A Heretical Theory' and 'Can Digital Computers Think?' and the panel discussion 'Can Automatic Calculating Machines be Said to Think?'[57]
c. The term was chosen by McCarthy to avoid associations with cybernetics and the influence of Norbert Wiener. "[O]ne of the reasons for inventing the term "artificial intelligence" was to escape association with "cybernetics". Its concentration on analog feedback seemed misguided, and I wished to avoid having either to accept Norbert (not Robert) Wiener as a guru or having to argue with him.".[77]
d. History would prove Moravec right about applications like computer vision. Moravec estimated that simply matching the edge and motion detection capabilities of the human retina in real time would require a general-purpose computer capable of 1000 million instructions per second (MIPS).[129] In 1976, the fastest supercomputer, the $8 million Cray-1 was only capable of 130 MIPS, and a typical desktop computer had 1 MIPS. As of 2011, practical computer vision applications require 10,000 to 1,000,000 MIPS.
e. John McCarthy wrote in response that "the combinatorial explosion problem has been recognized in AI from the beginning" in Review of Lighthill report (http://www-formal.stanford.edu/jmc/reviews/lighthill/lighthill.html)
f. This account is based on Crevier 1993, pp. 115–116. Other views include McCorduck 2004, pp. 306–313 and NRC 1999 under "Success in Speech Recognition".

g. Moravec explains, "Their initial promises to DARPA had been much too optimistic. Of course, what they delivered stopped considerably short of that. But they felt they couldn't in their next proposal promise less than in the first one, so they promised more."[138]

h. While the autonomous tank was a failure, the battle management system (called "DART") proved to be enormously successful, saving billions in the first Gulf War, repaying the investment and justifying the DARPA's pragmatic policy, at least as far as DARPA was concerned.

i. Versions of backpropagation had been developed in several fields (most directly as the reverse mode of automatic differentiation published by Seppo Linnainmaa (1970). It was applied to neural networks in the 1970s by Paul Werbos.

j. Hans Moravec wrote: "I am confident that this bottom-up route to artificial intelligence will one date meet the traditional top-down route more than half way, ready to provide the real world competence and the commonsense knowledge that has been so frustratingly elusive in reasoning programs. Fully intelligent machines will result when the metaphorical golden spike is driven uniting the two efforts."[182]

1. Kaplan A, Haenlein M (2019). "Siri, Siri, in my hand: Who's the fairest in the land? On the interpretations, illustrations, and implications of artificial intelligence". *Business Horizons*. **62**: 15–25. doi:10.1016/j.bushor.2018.08.004 (https://doi.org/10.1016%2Fj.bushor.2018.08.004). S2CID 158433736 (https://api.semanticscholar.org/CorpusID:158433736).
2. Newquist 1994, pp. 143–156.
3. Newquist 1994, pp. 144–152.
4. The Talos episode in Argonautica 4
5. Bibliotheke 1.9.26
6. Rhodios A (2007). *The Argonautika : Expanded Edition*. University of California Press. p. 355. ISBN 978-0-520-93439-9. OCLC 811491744 (https://www.worldcat.org/oclc/811491744).
7. Morford M (2007). *Classical mythology*. Oxford: Oxford University Press. p. 184. ISBN 978-0-19-085164-4. OCLC 1102437035 (https://www.worldcat.org/oclc/1102437035).
8. Linden SJ (2003). *The alchemy reader : from Hermes Trismegistus to Isaac Newton*. New York: Cambridge University Press. pp. Ch. 18. ISBN 0-521-79234-7. OCLC 51210362 (https://www.worldcat.org/oclc/51210362).
9. Kressel M (1 October 2015). "36 Days of Judaic Myth: Day 24, The Golem of Prague" (https://www.matthewkressel.net/2015/10/01/36-days-of-judaic-myth-day-24-the-golem-of-prague/). *Matthew Kressel*. Retrieved 15 March 2020.
10. Newquist 1994, p. .
11. "GOLEM" (http://www.jewishencyclopedia.com/articles/6777-golem). *www.jewishencyclopedia.com*. Retrieved 15 March 2020.
12. Newquist 1994, p. 38.
13. "Sanhedrin 65b" (https://www.sefaria.org/Sanhedrin.65b?lang=bi). *www.sefaria.org*. Retrieved 15 March 2020.
14. O'Connor KM (1994). "The alchemical creation of life (takwin) and other concepts of Genesis in medieval Islam" (https://repository.upenn.edu/dissertations/AAI9503804). *Dissertations Available from ProQuest*: 1–435.
15. Goethe JW (1890). *Faust; a tragedy. Translated, in the original metres ... by Bayard Taylor. Authorised ed., published by special arrangement with Mrs. Bayard Taylor. With a biographical introd* (https://archive.org/details/fausttragedytran00goetuoft). London Ward, Lock.
16. McCorduck 2004, pp. 17–25.
17. Butler 1863.
18. Newquist 1994, p. 65.
19. Cave S, Dihal K (2019). "Hopes and fears for intelligent machines in fiction and reality" (https://www.nature.com/articles/s42256-019-0020-9). *Nature Machine Intelligence*. **1** (2): 74–78. doi:10.1038/s42256-019-0020-9 (https://doi.org/10.1038%2Fs42256-019-0020-9). ISSN 2522-5839 (https://www.worldcat.org/issn/2522-5839). S2CID 150700981 (https://api.semanticscholar.org/CorpusID:150700981).
20. Needham 1986, p. 53.
21. McCorduck 2004, p. 6.
22. Nick 2005.
23. McCorduck 2004, p. 10.
24. Newquist 1994, p. 40.
25. McCorduck 2004, p. 16.
26. McCorduck 2004, pp. 59–62.

27. McCorduck 2004, p. 17.
28. Levitt 2000.
29. Newquist 1994, p. 30.
30. Crevier 1993, p. 1.
31. Quoted in McCorduck 2004, p. 8.
32. Cave S, Dihal K, Dillon S (2020). *AI Narratives: A History of Imaginative Thinking about Intelligent Machines* (https://books.google.com/books?id=T53SDwAAQBAJ&pg=PA56). Oxford University Press. p. 56. ISBN 978-0-19-884666-6. Retrieved 2 May 2023.
33. Butler, E. M. (Eliza Marian) (1948). *The myth of the magus*. London: Cambridge University Press. ISBN 0-521-22564-7. OCLC 5063114 (https://www.worldcat.org/oclc/5063114).
34. Porterfield A (2006). *The Protestant Experience in America* (https://books.google.com/books?id=V9VM9NEsqXwC&pg=PA136). American religious experience. Greenwood Press. p. 136. ISBN 978-0-313-32801-5. Retrieved 15 May 2023.
35. Hollander LM (1964). *Heimskringla; history of the kings of Norway*. Austin: Published for the American-Scandinavian Foundation by the University of Texas Press. ISBN 0-292-73061-6. OCLC 638953 (https://www.worldcat.org/oclc/638953).
36. Berlinski 2000.
37. Cfr. Carreras Artau, Tomás y Joaquín. *Historia de la filosofía española. Filosofía cristiana de los siglos XIII al XV*. Madrid, 1939, Volume I
38. Bonner, Anthonny, *The Art and Logic of Ramón Llull: A User's Guide*, Brill, 2007.
39. Anthony Bonner (ed.), Doctor Illuminatus. A Ramon Llull Reader (Princeton University 1985). Vid. "Llull's Influence: The History of Lullism" at 57–71
40. 17th century mechanism and AI:
    - McCorduck 2004, pp. 37–46
    - Russell & Norvig 2003, p. 6
    - Buchanan 2005, p. 53
41. Hobbes and AI:
    - McCorduck 2004, p. 42
    - Hobbes 1651, chapter 5
42. Leibniz and AI:
    - McCorduck 2004, p. 41
    - Russell & Norvig 2003, p. 6
    - Berlinski 2000, p. 12
    - Buchanan 2005, p. 53
43. Crevier 1993, pp. 190 196, 61.
44. The original photo can be seen in the article: Rose A (April 1946). "Lightning Strikes Mathematics" (https://books.google.com/books?id=niEDAAAAMBAJ&q=eniac+intitle:popular+intitle:science&pg=PA83). *Popular Science*: 83–86. Retrieved 15 April 2012.
45. The Turing machine: Newquist 1994, p. 56, McCorduck 2004, pp. 63–64, Crevier 1993, pp. 22–24, Russell & Norvig 2003, p. 8 and see Turing 1936–1937
46. Couturat 1901.
47. Russell & Norvig 2021, p. 15.
48. Russell & Norvig (2021, p. 15); Newquist (1994, p. 67)
49. Randall (1982, pp. 4–5); Byrne (2012); Mulvihill (2012)
50. Randall (1982, pp. 6, 11–13); Quevedo (1914); Quevedo (1915)
51. Randall 1982, pp. 13, 16–17.
52. Quoted in Russell & Norvig (2021, p. 15)
53. Menabrea & Lovelace 1843.
54. Russell & Norvig 2021, p. 14.
55. McCorduck 2004, pp. 76–80.
56. McCorduck 2004, pp. 51–57, 80–107, Crevier 1993, pp. 27–32, Russell & Norvig 2003, pp. 15, 940, Moravec 1988, p. 3, Cordeschi 2002, Chap. 5.
57. Copeland 2004.

58. McCorduck 2004, pp. 111–136, Crevier 1993, pp. 49–51, Russell & Norvig 2003, p. 17, Newquist 1994, pp. 91–112 and Kaplan A. "Artificial Intelligence, Business and Civilization - Our Fate Made in Machines" (https://www.routledge.com/Artificial-Intelligence-Business-and-Civilization-Our-Fate-Made-in-Machines/Kaplan/p/book/9781032155319). Retrieved 11 March 2022.
59. Image adapted from Saygin 2000
60. McCorduck 2004, pp. 70–72, Crevier 1993, pp. 22−25, Russell & Norvig 2003, pp. 2–3 and 948, Haugeland 1985, pp. 6–9, Cordeschi 2002, pp. 170–176. See also Turing 1950
61. Newquist 1994, pp. 92–98.
62. Russell & Norvig (2003, p. 948) claim that Turing answered all the major objections to AI that have been offered in the years since the paper appeared.
63. McCulloch WS, Pitts W (1 December 1943). "A logical calculus of the ideas immanent in nervous activity". *Bulletin of Mathematical Biophysics*. **5** (4): 115–133. doi:10.1007/BF02478259 (https://doi.org/10.1007%2FBF02478259). ISSN 1522-9602 (https://www.worldcat.org/issn/1522-9602).
64. Piccinini G (1 August 2004). "The First Computational Theory of Mind and Brain: A Close Look at Mcculloch and Pitts's "Logical Calculus of Ideas Immanent in Nervous Activity" ". *Synthese*. **141** (2): 175–215. doi:10.1023/B:SYNT.0000043018.52445.3e (https://doi.org/10.1023%2FB%3ASYNT.0000043018.52445.3e). ISSN 1573-0964 (https://www.worldcat.org/issn/1573-0964). S2CID 10442035 (https://api.semanticscholar.org/CorpusID:10442035).
65. McCorduck 2004, pp. 51–57, 88–94, Crevier 1993, p. 30, Russell & Norvig 2003, pp. 15−16, Cordeschi 2002, Chap. 5 and see also McCullough & Pitts 1943
66. McCorduck 2004, p. 102, Crevier 1993, pp. 34–35 and Russell & Norvig 2003, p. 17
67. McCorduck 2004, p. 98, Crevier 1993, pp. 27–28, Russell & Norvig 2003, pp. 15, 940, Moravec 1988, p. 3, Cordeschi 2002, Chap. 5.
68. See "A Brief History of Computing" (http://www.alanturing.net/turing_archive/pages/Reference%20Articles/BriefHistofComp.html) at AlanTuring.net.
69. Schaeffer, Jonathan. *One Jump Ahead:: Challenging Human Supremacy in Checkers*, 1997, 2009, Springer, ISBN 978-0-387-76575-4. Chapter 6.
70. McCorduck 2004, pp. 137–170, Crevier 1993, pp. 44–47
71. McCorduck 2004, pp. 123–125, Crevier 1993, pp. 44–46 and Russell & Norvig 2003, p. 17
72. Quoted in Crevier 1993, p. 46 and Russell & Norvig 2003, p. 17
73. Russell & Norvig 2003, pp. 947, 952
74. McCorduck 2004, pp. 111–136, Crevier 1993, pp. 49–51 and Russell & Norvig 2003, p. 17 Newquist 1994, pp. 91–112
75. See McCarthy et al. 1955. Also, see Crevier 1993, p. 48 where Crevier states "[the proposal] later became known as the 'physical symbol systems hypothesis'". The physical symbol system hypothesis was articulated and named by Newell and Simon in their paper on GPS. (Newell & Simon 1963) It includes a more specific definition of a "machine" as an agent that manipulates symbols. See the philosophy of artificial intelligence.
76. "I won't swear and I hadn't seen it before," McCarthy told Pamela McCorduck in 1979. (McCorduck 2004, p. 114) However, McCarthy also stated unequivocally "I came up with the term" in a CNET interview. (Skillings 2006)
77. McCarthy J (1988). "Review of *The Question of Artificial Intelligence*". *Annals of the History of Computing*. **10** (3): 224–229., collected in McCarthy J (1996). "10. Review of *The Question of Artificial Intelligence*". *Defending AI Research: A Collection of Essays and Reviews*. CSLI., p. 73
78. McCorduck (2004, pp. 129–130) discusses how the Dartmouth conference alumni dominated the first two decades of AI research, calling them the "invisible college".
79. McCorduck 2004, pp. 125.
80. Crevier (1993, pp. 49) writes "the conference is generally recognized as the official birthdate of the new science."
81. Miller G (2003). "The cognitive revolution: a historical perspective" (https://www.cs.princeton.edu/~rit/geo/Miller.pdf) (PDF). *Trends in Cognitive Sciences*. **7** (3): 141–144. doi:10.1016/s1364-6613(03)00029-9 (https://doi.org/10.1016%2Fs1364-6613%2803%2900029-9). PMID 12639696 (https://pubmed.ncbi.nlm.nih.gov/12639696).
82. Russell and Norvig write "it was astonishing whenever a computer did anything remotely clever." Russell & Norvig 2003, p. 18
83. Crevier 1993, pp. 52–107, Moravec 1988, p. 9 and Russell & Norvig 2003, pp. 18−21
84. McCorduck 2004, p. 218, Newquist 1994, pp. 91–112, Crevier 1993, pp. 108–109 and Russell & Norvig 2003, p. 21

85. Crevier 1993, pp. 52–107, Moravec 1988, p. 9
86. Heuristic: McCorduck 2004, p. 246, Russell & Norvig 2003, pp. 21–22
87. GPS: McCorduck 2004, pp. 245–250, Crevier 1993, p. GPS?, Russell & Norvig 2003, p. GPS?
88. Crevier 1993, pp. 51–58, 65–66 and Russell & Norvig 2003, pp. 18–19
89. McCorduck 2004, pp. 268–271, Crevier 1993, pp. 95–96, Newquist 1994, pp. 148–156, Moravec 1988, pp. 14–15
90. McCorduck 2004, p. 286, Crevier 1993, pp. 76–79, Russell & Norvig 2003, p. 19
91. Crevier 1993, pp. 79–83
92. Crevier 1993, pp. 164–172
93. McCorduck 2004, pp. 291–296, Crevier 1993, pp. 134–139
94. McCorduck 2004, pp. 299–305, Crevier 1993, pp. 83–102, Russell & Norvig 2003, p. 19 and Copeland 2000
95. McCorduck 2004, pp. 300–305, Crevier 1993, pp. 84–102, Russell & Norvig 2003, p. 19
96. Hecht-Nielsen R (1991). *Neurocomputing* (Reprint. with corrections ed.). Reading (Mass.) Menlo Park (Calif.) New York [etc.]: Addison-Wesley. p. 6, Figure 1.3 caption. ISBN 978-0-201-09355-1.
97. Block HD (1 January 1962). "The Perceptron: A Model for Brain Functioning. I" (https://link.aps.org/doi/10.1103/RevModPhys.34.123). *Reviews of Modern Physics*. **34** (1): 123–135. Bibcode:1962RvMP...34..123B (https://ui.adsabs.harvard.edu/abs/1962RvMP...34..123B). doi:10.1103/RevModPhys.34.123 (https://doi.org/10.1103%2FRevModPhys.34.123). ISSN 0034-6861 (https://www.worldcat.org/issn/0034-6861).
98. McCorduck 2004, pp. 104–107, Crevier 1993, pp. 102–105, Russell & Norvig 2003, p. 22
99. Rosenblatt, Frank. *Principles of neurodynamics: Perceptrons and the theory of brain mechanisms*. Vol. 55. Washington, DC: Spartan books, 1962.
100. Widrow B, Lehr M (September 1990). "30 years of adaptive neural networks: perceptron, Madaline, and backpropagation" (https://ieeexplore.ieee.org/document/58323). *Proceedings of the IEEE*. **78** (9): 1415–1442. doi:10.1109/5.58323 (https://doi.org/10.1109%2F5.58323). S2CID 195704643 (https://api.semanticscholar.org/CorpusID:195704643).
101. Rosen, Charles A., Nils J. Nilsson, and Milton B. Adams. "A research and development program in applications of intelligent automata to reconnaissance-phase I. (https://web.archive.org/web/20060316081320/http://www.ai.sri.com/pubs/files/rosen65-esu65-1tech.pdf)" Proposal for Research SRI No. ESU 65-1, 8 January 1965.
102. Nilsson, Nils J. *The SRI Artificial Intelligence Center: A Brief History (https://web.archive.org/web/20220810142945/https://www.sri.com/wp-content/uploads/2021/12/635.pdf)*. Artificial Intelligence Center, SRI International, 1984.
103. Hart PE, Nilsson NJ, Perrault R, Mitchell T, Kulikowski CA, Leake DB (15 March 2003). "In Memoriam: Charles Rosen, Norman Nielsen, and Saul Amarel" (https://ojs.aaai.org/aimagazine/index.php/aimagazine/article/view/1683). *AI Magazine*. **24** (1): 6. doi:10.1609/aimag.v24i1.1683 (https://doi.org/10.1609%2Faimag.v24i1.1683). ISSN 2371-9621 (https://www.worldcat.org/issn/2371-9621).
104. Nilsson 2009, Section 4.2: Neural Networks.
105. Nielson DL (1 January 2005). "Chapter 4: The Life and Times of a Successful SRI Laboratory: Artificial Intelligence and Robotics" (https://www.sri.com/wp-content/uploads/2022/08/A-heritage-of-innovation-The-Life-and-Times-of-a-Successful-SRI-Laboratory-Artificial-Intelligence-and-Robotics.pdf) (PDF). *A HERITAGE OF INNOVATION SRI's First Half Century* (https://www.sri.com/publication/a-heritage-of-innovation-sris-first-half-century/) (1st ed.). SRI International. ISBN 978-0-9745208-0-3.
106. Olazaran Rodriguez, Jose Miguel. *A historical sociology of neural network research (https://web.archive.org/web/20221111165150/https://era.ed.ac.uk/bitstream/handle/1842/20075/Olazaran-RodriguezJM_1991redux.pdf?sequence=1&isAllowed=y)*. PhD Dissertation. University of Edinburgh, 1991. See especially Chapter 2 and 3.
107. Minsky & Papert 1969.
108. Simon & Newell 1958, pp. 7−8 quoted in Crevier 1993, p. 108. See also Russell & Norvig 2003, p. 21
109. Simon 1965, p. 96 quoted in Crevier 1993, p. 109
110. Minsky 1967, p. 2 quoted in Crevier 1993, p. 109
111. Minsky strongly believes he was misquoted. See McCorduck 2004, pp. 272–274, Crevier 1993, p. 96 and Darrach 1970.
112. Crevier 1993, pp. 64–65
113. Crevier 1993, p. 94
114. Howe 1994

115. McCorduck 2004, p. 131, Crevier 1993, p. 51. McCorduck also notes that funding was mostly under the direction of alumni of the Dartmouth workshop of 1956.
116. Crevier 1993, p. 65
117. Crevier 1993, pp. 68–71 and Turkle 1984
118. Crevier 1993, pp. 163–196.
119. Dreyfus 1972.
120. Lighthill 1973.
121. Haigh 2023.
122. Crevier 1993, p. 143.
123. Nilsson 2009, p. 1.
124. Crevier 1993, p. 146
125. Russell & Norvig 2003, pp. 20–21Newquist 1994, pp. 336
126. Buchanan 2005, p. 56: "Early programs were necessarily limited in scope by the size and speed of memory"
127. Crevier 1993, pp. 146–148.
128. Moravec 1976.
129. Hans Moravec, *ROBOT: Mere Machine to Transcendent Mind*
130. Russell & Norvig 2003, pp. 9, 21–22 and Lighthill 1973
131. McCorduck 2004, pp. 300 & 421; Crevier 1993, pp. 113–114; Moravec 1988, p. 13; Lenat & Guha 1989, (Introduction); Russell & Norvig 2003, p. 21
132. McCorduck 2004, p. 456, Moravec 1988, pp. 15–16
133. McCarthy & Hayes 1969, Crevier 1993, pp. 117–119
134. McCorduck 2004, pp. 280–281; Crevier 1993, p. 110; Russell & Norvig 2003, p. 21; NRC 1999, under "Success in Speech Recognition".
135. Crevier 1993, p. 117; Russell & Norvig 2003, p. 22; Howe 1994 Lighthill 1973
136. Russell & Norvig 2003, p. 22; Lighthill 1973
137. Crevier 1993, pp. 115–116.
138. Crevier 1993, p. 115.
139. NRC 1999, under "Shift to Applied Research Increases Investment.".
140. Lucas and Penrose' critique of AI: Crevier 1993, p. 22, Russell & Norvig 2003, pp. 949–950, Hofstadter 1999, pp. 471–477 and see Lucas 1961
141. "Know-how" is Dreyfus' term. (Dreyfus makes a distinction between "knowing how" and "knowing that", a modern version of Heidegger's distinction of ready-to-hand and present-at-hand.) (Dreyfus & Dreyfus 1986)
142. Dreyfus' critique of artificial intelligence: McCorduck 2004, pp. 211–239, Crevier 1993, pp. 120–132, Russell & Norvig 2003, pp. 950–952 and see Dreyfus 1965, Dreyfus 1972, Dreyfus & Dreyfus 1986
143. Searle's critique of AI: McCorduck 2004, pp. 443–445, Crevier 1993, pp. 269–271, Russell & Norvig 2003, pp. 958–960 and see Searle 1980
144. Quoted in Crevier 1993, p. 143
145. Quoted in Crevier 1993, p. 122
146. "I became the only member of the AI community to be seen eating lunch with Dreyfus. And I deliberately made it plain that theirs was not the way to treat a human being." Joseph Weizenbaum, quoted in Crevier 1993, p. 123.
147. Newquist 1994, pp. 276
148. Colby, Watt & Gilbert 1966, p. 148. Weizenbaum referred to this text in Weizenbaum 1976, pp. 5, 6. Colby and his colleagues later also developed chatterbot-like "computer simulations of paranoid processes (PARRY)" to "make intelligible paranoid processes in explicit symbol processing terms." (Colby 1974, p. 6)
149. Weizenbaum's critique of AI: McCorduck 2004, pp. 356–373, Crevier 1993, pp. 132–144, Russell & Norvig 2003, p. 961 and see Weizenbaum 1976
150. McCorduck 2004, p. 51, Russell & Norvig 2003, pp. 19, 23
151. McCorduck 2004, p. 51, Crevier 1993, pp. 190–192
152. Crevier 1993, pp. 193–196
153. Crevier 1993, pp. 145–149, 258–63

154. Wason & Shapiro (1966) showed that people do poorly on completely abstract problems, but if the problem is restated to allow the use of intuitive social intelligence, performance dramatically improves. (See Wason selection task) Kahneman, Slovic & Tversky (1982) have shown that people are terrible at elementary problems that involve uncertain reasoning. (See list of cognitive biases for several examples). Eleanor Rosch's work is described in Lakoff 1987
155. An early example of McCarthy's position was in the journal Science where he said "This is AI, so we don't care if it's psychologically real" (Kolata 1982), and he recently reiterated his position at the AI@50 conference where he said "Artificial intelligence is not, by definition, simulation of human intelligence" (Maker 2006).
156. Crevier 1993, pp. 175
157. Neat vs. scruffy: McCorduck 2004, pp. 421–424 (who picks up the state of the debate in 1984). Crevier 1993, pp. 168 (who documents Schank's original use of the term). Another aspect of the conflict was called "the procedural/declarative distinction" but did not prove to be influential in later AI research.
158. McCorduck 2004, pp. 305–306, Crevier 1993, pp. 170–173, 246 and Russell & Norvig 2003, p. 24. Minsky's frame paper: Minsky 1974.
159. Hayes P (1981). "The logic of frames". In Kaufmann M (ed.). *Readings in artificial intelligence*. pp. 451–458.
160. Reiter R (1978). "On reasoning by default". *American Journal of Computational Linguistics*: 29–37.
161. Clark K (1977). "Negation as Failure". *Logic and Data Bases*. Boston, MA: Springer US. pp. 293–322. doi:10.1007/978-1-4684-3384-5_11 (https://doi.org/10.1007%2F978-1-4684-3384-5_11). ISBN 978-1-4684-3386-9.
162. McCorduck 2004, pp. 327–335 (Dendral), Crevier 1993, pp. 148–159, Newquist 1994, p. 271, Russell & Norvig 2003, pp. 22–23
163. Crevier 1993, pp. 158–159 and Russell & Norvig 2003, pp. 23−24
164. Crevier 1993, p. 198
165. Newquist 1994, pp. 259
166. McCorduck 2004, pp. 434–435, Crevier 1993, pp. 161–162, 197–203, Newquist 1994, pp. 275 and Russell & Norvig 2003, p. 24
167. McCorduck 2004, pp. 436–441, Newquist 1994, pp. 231–240, Crevier 1993, pp. 211, Russell & Norvig 2003, p. 24 and see also Feigenbaum & McCorduck 1983
168. Crevier 1993, pp. 195.
169. Crevier 1993, pp. 240.
170. Russell & Norvig 2003, p. 25.
171. McCorduck 2004, pp. 426–432, NRC 1999, under "Shift to Applied Research Increases Investment"
172. McCorduck 2004, p. 299
173. McCorduck 2004, pp. 421
174. Knowledge revolution: McCorduck 2004, pp. 266–276, 298–300, 314, 421, Newquist 1994, pp. 255–267, Russell & Norvig 2003, pp. 22–23
175. Cyc: McCorduck 2004, p. 489, Crevier 1993, pp. 239–243, Newquist 1994, pp. 431–455, Russell & Norvig 2003, pp. 363−365 and Lenat & Guha 1989
176. "Chess: Checkmate" (https://web.archive.org/web/20071008204404/http://www.computerhistory.org/about/press_relations/media_kit/chess/5_chess_exhibit_text_Endgame.pdf) (PDF). Archived from the original (http://www.computerhistory.org/about/press_relations/media_kit/chess/5_chess_exhibit_text_Endgame.pdf) (PDF) on 8 October 2007. Retrieved 1 September 2007.
177. Sejnowski TJ (23 October 2018). *The Deep Learning Revolution* (1st ed.). Cambridge, Massachusetts London, England: The MIT Press. pp. 93–94. ISBN 978-0-262-03803-4.
178. Crevier 1993, pp. 214–215.
179. Russell & Norvig 2021, p. 26.
180. Christian 2020, pp. 21–22.
181. McCorduck 2004, pp. 454–462.
182. Moravec 1988, p. 20.
183. Crevier 1993, pp. 183–190.
184. Brooks 1990.
185. Brooks 1990, p. 3.
186. See, for example, Lakoff & Johnson 1999
187. Newquist 1994, pp. 501, 511.

188. McCorduck 2004, p. 424.
189. Crevier 1993, pp. 203. AI winter was first used as the title of a seminar on the subject for the Association for the Advancement of Artificial Intelligence.
190. Newquist 1994, pp. 359–379, McCorduck 2004, p. 435, Crevier 1993, pp. 209–210
191. McCorduck 2004, p. 435 (who cites institutional reasons for their ultimate failure), Newquist 1994, pp. 258–283 (who cites limited deployment within corporations), Crevier 1993, pp. 204–208 (who cites the difficulty of truth maintenance, i.e., learning and updating), Lenat & Guha 1989, Introduction (who emphasizes the brittleness and the inability to handle excessive qualification.)
192. McCorduck 2004, pp. 430–431
193. McCorduck 2004, p. 441, Crevier 1993, p. 212. McCorduck writes "Two and a half decades later, we can see that the Japanese didn't quite meet all of those ambitious goals."
194. Newquist 1994, pp. 476
195. Newquist 1994, pp. 440
196. See Applications of artificial intelligence § Computer science
197. NRC 1999 under "Artificial Intelligence in the 90s", and Kurzweil 2005, p. 264
198. Russell & Norvig 2003, p. 28
199. For the new state of the art in AI based speech recognition, see The Economist (2007)
200. "AI-inspired systems were already integral to many everyday technologies such as internet search engines, bank software for processing transactions and in medical diagnosis." Nick Bostrom, quoted in CNN 2006
201. Olsen (2004),Olsen (2006)
202. McCorduck 2004, p. 423, Kurzweil 2005, p. 265, Hofstadter 1999, p. 601 Newquist 1994, pp. 445
203. CNN 2006
204. Markoff 2005
205. The Economist 2007
206. Tascarella 2006
207. Newquist 1994, pp. 532
208. McCorduck 2004, pp. 480–483
209. "Deep Blue" (http://www.research.ibm.com/deepblue/meet/html/d.3.shtml). IBM Research. Retrieved 10 September 2010.
210. "DARPA Grand Challenge – home page" (https://web.archive.org/web/20071031110030/http://www.darpa.mil/grandchallenge/). Archived from the original (https://www.darpa.mil/grandchallenge/) on 31 October 2007.
211. "Welcome" (https://web.archive.org/web/20140305023802/http://archive.darpa.mil/grandchallenge/). Archived from the original (https://archive.darpa.mil/grandchallenge/) on 5 March 2014. Retrieved 25 October 2011.
212. Markoff J (16 February 2011). "On 'Jeopardy!' Watson Win Is All but Trivial" (https://www.nytimes.com/2011/02/17/science/17jeopardy-watson.html). *The New York Times*.
213. Kurzweil 2005, p. 274 writes that the improvement in computer chess, "according to common wisdom, is governed only by the brute force expansion of computer hardware."
214. Cycle time of Ferranti Mark 1 was 1.2 milliseconds, which is arguably equivalent to about 833 flops. Deep Blue ran at 11.38 gigaflops (and this does not even take into account Deep Blue's special-purpose hardware for chess). *Very* approximately, these differ by a factor of 10^7.
215. McCorduck 2004, pp. 471–478, Russell & Norvig 2003, p. 55, where they write: "The whole-agent view is now widely accepted in the field". The intelligent agent paradigm is discussed in major AI textbooks, such as: Russell & Norvig 2003, pp. 32–58, 968–972, Poole, Mackworth & Goebel 1998, pp. 7–21, Luger & Stubblefield 2004, pp. 235–240
216. Carl Hewitt's Actor model anticipated the modern definition of intelligent agents. (Hewitt, Bishop & Steiger 1973) Both John Doyle (Doyle 1983) and Marvin Minsky's popular classic *The Society of Mind* (Minsky 1986) used the word "agent". Other "modular" proposals included Rodney Brook's subsumption architecture, object-oriented programming and others.
217. Russell & Norvig 2003, pp. 27, 55
218. This is how the most widely accepted textbooks of the 21st century define artificial intelligence. See Russell & Norvig 2003, p. 32 and Poole, Mackworth & Goebel 1998, p. 1
219. McCorduck 2004, p. 478
220. McCorduck 2004, pp. 486–487, Russell & Norvig 2003, pp. 25–26

221. Pearl 1988
222. Russell & Norvig 2003, pp. 25−26
223. Steve Lohr (17 October 2016), "IBM Is Counting on Its Bet on Watson, and Paying Big Money for It" (https://www.nytimes.com/2016/10/17/technology/ibm-is-counting-on-its-bet-on-watson-and-paying-big-money-for-it.html?emc=edit_th_20161017&nl=todaysheadlines&nlid=62816440), *New York Times*
224. Hampton SE, Strasser CA, Tewksbury JJ, Gram WK, Budden AE, Batcheller AL, Duke CS, Porter JH (1 April 2013). "Big data and the future of ecology" (http://www.escholarship.org/uc/item/94f35801). *Frontiers in Ecology and the Environment*. **11** (3): 156–162. Bibcode:2013FrEE...11..156H (https://ui.adsabs.harvard.edu/abs/2013FrEE...11..156H). doi:10.1890/120103 (https://doi.org/10.1890%2F120103). ISSN 1540-9309 (https://www.worldcat.org/issn/1540-9309).
225. "How Big Data is Changing Economies | Becker Friedman Institute" (https://web.archive.org/web/20180618102343/https://bfi.uchicago.edu/events/how-big-data-changing-economies). *bfi.uchicago.edu*. Archived from the original (https://bfi.uchicago.edu/events/how-big-data-changing-economies) on 18 June 2018. Retrieved 9 June 2017.
226. LeCun Y, Bengio Y, Hinton G (2015). "Deep learning" (https://hal.science/hal-04206682/file/Lecun2015.pdf) (PDF). *Nature*. **521** (7553): 436–444. Bibcode:2015Natur.521..436L (https://ui.adsabs.harvard.edu/abs/2015Natur.521..436L). doi:10.1038/nature14539 (https://doi.org/10.1038%2Fnature14539). PMID 26017442 (https://pubmed.ncbi.nlm.nih.gov/26017442). S2CID 3074096 (https://api.semanticscholar.org/CorpusID:3074096).
227. Milmo D (3 November 2023). "Hope or Horror? The great AI debate dividing its pioneers". *The Guardian Weekly*. pp. 10–12.
228. "The Bletchley Declaration by Countries Attending the AI Safety Summit, 1-2 November 2023" (https://web.archive.org/web/20231101123904/https://www.gov.uk/government/publications/ai-safety-summit-2023-the-bletchley-declaration/the-bletchley-declaration-by-countries-attending-the-ai-safety-summit-1-2-november-2023). *GOV.UK*. 1 November 2023. Archived from the original (https://www.gov.uk/government/publications/ai-safety-summit-2023-the-bletchley-declaration/the-bletchley-declaration-by-countries-attending-the-ai-safety-summit-1-2-november-2023) on 1 November 2023. Retrieved 2 November 2023.
229. "Countries agree to safe and responsible development of frontier AI in landmark Bletchley Declaration" (https://www.gov.uk/government/news/countries-agree-to-safe-and-responsible-development-of-frontier-ai-in-landmark-bletchley-declaration). *GOV.UK* (Press release). Archived (https://web.archive.org/web/20231101115016/https://www.gov.uk/government/news/countries-agree-to-safe-and-responsible-development-of-frontier-ai-in-landmark-bletchley-declaration) from the original on 1 November 2023. Retrieved 1 November 2023.
230. Baral C, Fuentes O, Kreinovich V (June 2015). "Why Deep Neural Networks: A Possible Theoretical Explanation" (http://digitalcommons.utep.edu/cs_techrep/975/). *Departmental Technical Reports (Cs)*. Retrieved 9 June 2017.
231. Ciregan D, Meier U, Schmidhuber J (June 2012). "Multi-column deep neural networks for image classification". *2012 IEEE Conference on Computer Vision and Pattern Recognition*. pp. 3642–3649. arXiv:1202.2745 (https://arxiv.org/abs/1202.2745). Bibcode:2012arXiv1202.2745C (https://ui.adsabs.harvard.edu/abs/2012arXiv1202.2745C). CiteSeerX 10.1.1.300.3283 (https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.300.3283). doi:10.1109/cvpr.2012.6248110 (https://doi.org/10.1109%2Fcvpr.2012.6248110). ISBN 978-1-4673-1228-8. S2CID 2161592 (https://api.semanticscholar.org/CorpusID:2161592).
232. Markoff J (16 February 2011). "On 'Jeopardy!' Watson Win Is All but Trivial" (https://www.nytimes.com/2011/02/17/science/17jeopardy-watson.html). *The New York Times*. ISSN 0362-4331 (https://www.worldcat.org/issn/0362-4331). Retrieved 10 June 2017.
233. "AlphaGo: Mastering the ancient game of Go with Machine Learning" (https://research.googleblog.com/2016/01/alphago-mastering-ancient-game-of-go.html). *Research Blog*. Retrieved 10 June 2017.
234. "Innovations of AlphaGo | DeepMind" (https://deepmind.com/blog/innovations-alphago/). *DeepMind*. 10 April 2017. Retrieved 10 June 2017.
235. University CM. "Computer Out-Plays Humans in "Doom"-CMU News - Carnegie Mellon University" (https://www.cmu.edu/news/stories/archives/2016/september/AI-agent-survives-doom.html). *www.cmu.edu*. Retrieved 10 June 2017.
236. Laney D (2001). "3D data management: Controlling data volume, velocity and variety". *META Group Research Note*. **6** (70).
237. Marr, Bernard (6 March 2014). "Big Data: The 5 Vs Everyone Must Know" (https://www.linkedin.com/pulse/20140306073407-64875646-big-data-the-5-vs-everyone-must-know/).
238. Goes PB (2014). "Design science research in top information systems journals". *MIS Quarterly: Management Information Systems*. **38** (1).

239. Marr B. "Beyond The Hype: What You Really Need To Know About AI In 2023" (https://www.forbes.com/sites/bernardmarr/2023/03/20/beyond-the-hype-what-you-really-need-to-know-about-ai-in-2023/). *Forbes*. Retrieved 27 January 2024.
240. "The Era of AI: 2023's Landmark Year" (https://www.cmswire.com/digital-experience/the-era-of-ai-end-of-year-ai-recap/). *CMSWire.com*. Retrieved 28 January 2024.
241. "How the era of artificial intelligence will transform society?" (https://pocketconfidant.com/how-the-era-of-artificial-intelligence-will-transform-society/). *PocketConfidant AI*. 15 June 2018. Retrieved 28 January 2024.
242. "This year signaled the start of a new era" (https://www.linkedin.com/pulse/year-signaled-start-new-era-bill-gates-qbpfc). *www.linkedin.com*. Retrieved 28 January 2024.
243. Lee A (23 January 2024). "UT Designates 2024 'The Year of AI' " (https://news.utexas.edu/2024/01/23/ut-designates-2024-the-year-of-ai/). *UT News*. Retrieved 28 January 2024.
244. Murgia M (23 July 2023). "Transformers: the Google scientists who pioneered an AI revolution" (https://www.ft.com/content/37bb01af-ee46-4483-982f-ef3921436a50). *www.ft.com*. Retrieved 10 December 2023.
245. Bubeck S, Chandrasekaran V, Eldan R, Gehrke J, Horvitz E, Kamar E, Lee P, Lee YT, Li Y, Lundberg S, Nori H, Palangi H, Ribeiro MT, Zhang Y (22 March 2023). "Sparks of Artificial General Intelligence: Early experiments with GPT-4". arXiv:2303.12712 (https://arxiv.org/abs/2303.12712) [cs.CL (https://arxiv.org/archive/cs.CL)].

References

- Berlinski D (2000), *The Advent of the Algorithm* (https://archive.org/details/adventofalgorith0000berl), Harcourt Books, ISBN 978-0-15-601391-8, OCLC 46890682 (https://www.worldcat.org/oclc/46890682).
- Brooks RA (1990). "Elephants Don't Play Chess" (http://people.csail.mit.edu/brooks/papers/elephants.pdf) (PDF). *Robotics and Autonomous Systems*. **6** (1–2): 3–15. doi:10.1016/S0921-8890(05)80025-9 (https://doi.org/10.1016%2FS0921-8890%2805%2980025-9).
- Buchanan BG (Winter 2005), "A (Very) Brief History of Artificial Intelligence" (https://web.archive.org/web/20070926023314/http://www.aaai.org/AITopics/assets/PDF/AIMag26-04-016.pdf) (PDF), *AI Magazine*, pp. 53–60, archived from the original (http://www.aaai.org/AITopics/assets/PDF/AIMag26-04-016.pdf) (PDF) on 26 September 2007, retrieved 30 August 2007.
- Butler S (13 June 1863), "Darwin Among the Machines" (https://nzetc.victoria.ac.nz/tm/scholarly/tei-ButFir-t1-g1-t1-g1-t4-body.html), *The Press, Christchurch, New Zealand*, retrieved 10 October 2008.
- Byrne JG (8 December 2012). "The John Gabriel Byrne Computer Science Collection" (https://web.archive.org/web/20190416071721/https://www.scss.tcd.ie/SCSSTreasuresCatalog/miscellany/TCD-SCSS-X.20121208.002/TCD-SCSS-X.20121208.002.pdf) (PDF). Archived from the original (https://scss.tcd.ie/SCSSTreasuresCatalog/miscellany/TCD-SCSS-X.20121208.002/TCD-SCSS-X.20121208.002.pdf/) on 16 April 2019. Retrieved 8 August 2019.
- "AI set to exceed human brain power" (http://www.cnn.com/2006/TECH/science/07/24/ai.bostrom/), *CNN.com*, 26 July 2006, retrieved 16 October 2007.
- Colby KM, Watt JB, Gilbert JP (1966), "A Computer Method of Psychotherapy: Preliminary Communication" (https://exhibits.stanford.edu/feigenbaum/catalog/hk334rq4790), *The Journal of Nervous and Mental Disease*, vol. 142, no. 2, pp. 148–152, doi:10.1097/00005053-196602000-00005 (https://doi.org/10.1097%2F00005053-196602000-00005), PMID 5936301 (https://pubmed.ncbi.nlm.nih.gov/5936301), S2CID 36947398 (https://api.semanticscholar.org/CorpusID:36947398).
- Colby KM (September 1974), *Ten Criticisms of Parry* (http://i.stanford.edu/pub/cstr/reports/cs/tr/74/457/CS-TR-74-457.pdf) (PDF), Stanford Artificial Intelligence Laboratory, REPORT NO. STAN-CS-74-457, retrieved 17 June 2018.
- Couturat L (1901), *La Logique de Leibniz*
- Copeland J (2000), *Micro-World AI* (http://www.alanturing.net/turing_archive/pages/Reference%20Articles/what_is_AI/What%20is%20AI06.html), retrieved 8 October 2008.
- Copeland J((2004). *The Essential Turing: the ideas that gave birth to the computer age*. Oxford: Clarendon Press. ISBN 0-19-825079-7..
- Cordeschi R (2002), *The Discovery of the Artificial*, Dordrecht: Kluwer..
- Crevier D (1993). *AI: The Tumultuous Search for Artificial Intelligence*. New York, NY: BasicBooks. ISBN 0-465-02997-3.
- Darrach B (20 November 1970), "Meet Shaky, the First Electronic Person", *Life Magazine*, pp. 58–68.
- Doyle J (1983), "What is rational psychology? Toward a modern mental philosophy", *AI Magazine*, vol. 4, no. 3, pp. 50–53.

- Dreyfus H (1965), *Alchemy and AI*, RAND Corporation Memo.
- Dreyfus H (1972), *What Computers Can't Do*, New York: MIT Press, ISBN 978-0-06-090613-9, OCLC 5056816 (https://www.worldcat.org/oclc/5056816).
- Dreyfus H, Dreyfus S (1986). *Mind over Machine: The Power of Human Intuition and Expertise in the Era of the Computer* (https://archive.org/details/mindovermachinep00drey). Oxford, UK: Blackwell. ISBN 978-0-02-908060-3. Retrieved 22 August 2020.
- The Economist (7 June 2007), "Are You Talking to Me?" (http://www.economist.com/science/tq/displaystory.cfm?story_id=9249338), *The Economist*, retrieved 16 October 2008.
- Feigenbaum EA, McCorduck P (1983), *The Fifth Generation: Artificial Intelligence and Japan's Computer Challenge to the World*, Michael Joseph, ISBN 978-0-7181-2401-4.
- Haigh T (December 2023). "There Was No 'First AI Winter' " (https://dl.acm.org/doi/10.1145/3625833). *Communications of the ACM*. **66** (12): 35–39. doi:10.1145/3625833 (https://doi.org/10.1145%2F3625833). ISSN 0001-0782 (https://www.worldcat.org/issn/0001-0782)..
- Haugeland J (1985). *Artificial Intelligence: The Very Idea*. Cambridge, Mass.: MIT Press. ISBN 978-0-262-08153-5.
- Hawkins J, Blakeslee S (2004), *On Intelligence*, New York, NY: Owl Books, ISBN 978-0-8050-7853-4, OCLC 61273290 (https://www.worldcat.org/oclc/61273290).
- Hebb D (1949), *The Organization of Behavior*, New York: Wiley, ISBN 978-0-8058-4300-2, OCLC 48871099 (https://www.worldcat.org/oclc/48871099).
- Hewitt C, Bishop P, Steiger R (1973), *A Universal Modular Actor Formalism for Artificial Intelligence* (https://web.archive.org/web/20091229084457/http://dli.iiit.ac.in/ijcai/IJCAI-73/PDF/027B.pdf) (PDF), IJCAI, archived from the original (http://dli.iiit.ac.in/ijcai/IJCAI-73/PDF/027B.pdf) (PDF) on 29 December 2009
- Hobbes T (1651), *Leviathan*.
- Hofstadter D (1999) [1979], *Gödel, Escher, Bach: an Eternal Golden Braid*, Basic Books, ISBN 978-0-465-02656-2, OCLC 225590743 (https://www.worldcat.org/oclc/225590743).
- Howe J (November 1994), *Artificial Intelligence at Edinburgh University: a Perspective* (http://www.inf.ed.ac.uk/about/AIhistory.html), retrieved 30 August 2007.
- Kahneman D, Slovic D, Tversky A (1982). "Judgment under uncertainty: Heuristics and biases". *Science*. **185** (4157). New York: Cambridge University Press: 1124–1131. Bibcode:1974Sci...185.1124T (https://ui.adsabs.harvard.edu/abs/1974Sci...185.1124T). doi:10.1126/science.185.4157.1124 (https://doi.org/10.1126%2Fscience.185.4157.1124). ISBN 978-0-521-28414-1. PMID 17835457 (https://pubmed.ncbi.nlm.nih.gov/17835457). S2CID 143452957 (https://api.semanticscholar.org/CorpusID:143452957).
- Kaplan A, Haenlein M (2018), "Siri, Siri in my Hand, who's the Fairest in the Land? On the Interpretations, Illustrations and Implications of Artificial Intelligence", *Business Horizons*, **62**: 15–25, doi:10.1016/j.bushor.2018.08.004 (https://doi.org/10.1016%2Fj.bushor.2018.08.004), S2CID 158433736 (https://api.semanticscholar.org/CorpusID:158433736).
- Kolata G (1982), "How can computers get common sense?", *Science*, **217** (4566): 1237–1238, Bibcode:1982Sci...217.1237K (https://ui.adsabs.harvard.edu/abs/1982Sci...217.1237K), doi:10.1126/science.217.4566.1237 (https://doi.org/10.1126%2Fscience.217.4566.1237), PMID 17837639 (https://pubmed.ncbi.nlm.nih.gov/17837639).
- Kurzweil R (2005), *The Singularity is Near*, Viking Press, ISBN 978-0-14-303788-0, OCLC 71826177 (https://www.worldcat.org/oclc/71826177).
- Lakoff G (1987), *Women, Fire, and Dangerous Things: What Categories Reveal About the Mind* (https://archive.org/details/womenfiredangero00lako_0), University of Chicago Press., ISBN 978-0-226-46804-4.
- Lakoff G, Johnson M (1999). *Philosophy in the flesh: The embodied mind and its challenge to western thought* (https://www.basicbooks.com/titles/george-lakoff/philosophy-in-the-flesh/9780465056743/). Basic Books. ISBN 978-0-465-05674-3.
- Lenat D, Guha RV (1989), *Building Large Knowledge-Based Systems*, Addison-Wesley, ISBN 978-0-201-51752-1, OCLC 19981533 (https://www.worldcat.org/oclc/19981533).
- Levitt GM (2000), *The Turk, Chess Automaton*, Jefferson, N.C.: McFarland, ISBN 978-0-7864-0778-1.
- Lighthill PS (1973), "Artificial Intelligence: A General Survey", *Artificial Intelligence: a paper symposium*, Science Research Council
- Lucas J (1961), "Minds, Machines and Gödel", *Philosophy*, **36** (XXXVI): 112–127, doi:10.1017/S0031819100057983 (https://doi.org/10.1017%2FS0031819100057983), S2CID 55408480 (https://api.semanticscholar.org/CorpusID:55408480)
- Luger G, Stubblefield W (2004). *Artificial Intelligence: Structures and Strategies for Complex Problem Solving* (https://archive.org/details/artificialintell0000luge) (5th ed.). Benjamin/Cummings. ISBN 978-0-8053-4780-7. Retrieved 17 December 2019.

- Maker MH (2006), *AI@50: AI Past, Present, Future* (https://web.archive.org/web/20081008120238/http://www.engagingexperience.com/2006/07/ai50_ai_past_pr.html), Dartmouth College, archived from the original (http://www.engagingexperience.com/2006/07/ai50_ai_past_pr.html) on 8 October 2008, retrieved 16 October 2008
- Markoff J (14 October 2005), "Behind Artificial Intelligence, a Squadron of Bright Real People" (https://www.nytimes.com/2005/10/14/technology/14artificial.html?_r=1&ei=5070&en=11ab55edb7cead5e&ex=1185940800&adxnnl=1&adxnnlx=1185805173-o7WsfW7qaP0x5/NUs1cQCQ&oref=slogin), *The New York Times*, retrieved 16 October 2008
- McCarthy J, Minsky M, Rochester N, Shannon C (31 August 1955), *A Proposal for the Dartmouth Summer Research Project on Artificial Intelligence* (https://web.archive.org/web/20080930164306/http://www-formal.stanford.edu/jmc/history/dartmouth/dartmouth.html), archived from the original (http://www-formal.stanford.edu/jmc/history/dartmouth/dartmouth.html) on 30 September 2008, retrieved 16 October 2008
- McCarthy J, Hayes PJ (1969), "Some philosophical problems from the standpoint of artificial intelligence", in Meltzer BJ, Mitchie D (eds.), *Machine Intelligence 4* (http://www-formal.stanford.edu/jmc/mcchay69/mcchay69.html), Edinburgh University Press, pp. 463–502, retrieved 16 October 2008
- McCorduck P (2004), *Machines Who Think* (2nd ed.), Natick, MA: A. K. Peters, Ltd., ISBN 978-1-56881-205-2, OCLC 52197627 (https://www.worldcat.org/oclc/52197627).
- McCullough WS, Pitts W (1943), "A logical calculus of the ideas immanent in nervous activity", *Bulletin of Mathematical Biophysics*, **5** (4): 115–127, doi:10.1007/BF02478259 (https://doi.org/10.1007%2FBF02478259)
- Menabrea LF, Lovelace A (1843), "Sketch of the Analytical Engine Invented by Charles Babbage" (http://www.fourmilab.ch/babbage/sketch.html), *Scientific Memoirs*, **3**, retrieved 29 August 2008 With notes upon the Memoir by the Translator
- Minsky M (1967), *Computation: Finite and Infinite Machines*, Englewood Cliffs, N.J.: Prentice-Hall
- Minsky M, Papert S (1969), *Perceptrons: An Introduction to Computational Geometry* (https://archive.org/details/perceptronsintro00mins), The MIT Press, ISBN 978-0-262-63111-2, OCLC 16924756 (https://www.worldcat.org/oclc/16924756)
- Minsky M (1974), *A Framework for Representing Knowledge* (https://web.archive.org/web/20210107162402/http://web.media.mit.edu/~minsky/papers/Frames/frames.html), archived from the original (http://web.media.mit.edu/~minsky/papers/Frames/frames.html) on 7 January 2021, retrieved 16 October 2008
- Minsky M (1986), *The Society of Mind*, Simon and Schuster, ISBN 978-0-671-65713-0, OCLC 223353010 (https://www.worldcat.org/oclc/223353010)
- Minsky M (2001), *It's 2001. Where Is HAL?* (http://www.ddj.com/hpc-high-performance-computing/197700454?cid=RSSfeed_DDJ_AI), Dr. Dobb's Technetcast, retrieved 8 August 2009
- Moor J, ed. (2003), *The Turing Test: The Elusive Standard of Artificial Intelligence*, Dordrecht: Kluwer Academic Publishers, ISBN 978-1-4020-1205-1
- Moravec H (1976), *The Role of Raw Power in Intelligence* (https://web.archive.org/web/20160303232511/http://www.frc.ri.cmu.edu/users/hpm/project.archive/general.articles/1975/Raw.Power.html), archived from the original (http://www.frc.ri.cmu.edu/users/hpm/project.archive/general.articles/1975/Raw.Power.html) on 3 March 2016, retrieved 16 October 2008
- Moravec H (1988), *Mind Children* (https://archive.org/details/mindchildren00hans), Harvard University Press, ISBN 978-0-674-57618-6, OCLC 245755104 (https://www.worldcat.org/oclc/245755104)
- Mulvihill M (17 October 2012). "1907: was the first portable computer design Irish?" (http://ingeniousireland.ie/2012/10/1909-a-novel-irish-computer/). *Ingenious Ireland*.
- Needham J (1986). *Science and Civilization in China: Volume 2*. Taipei: Caves Books Ltd.
- Newell A, Simon HA (1963), "GPS: A Program that Simulates Human Thought", in Feigenbaum E, Feldman J (eds.), *Computers and Thought*, New York: McGraw-Hill, ISBN 978-0-262-56092-4, OCLC 246968117 (https://www.worldcat.org/oclc/246968117)
- Newquist HP (1994), *The Brain Makers: Genius, Ego, And Greed in the Quest For Machines That Think*, New York: Macmillan/SAMS, ISBN 978-0-9885937-1-8, OCLC 313139906 (https://www.worldcat.org/oclc/313139906)
- NRC (1999), "Developments in Artificial Intelligence" (https://archive.org/details/fundingrevolutio00nati), *Funding a Revolution: Government Support for Computing Research*, National Academy Press, ISBN 978-0-309-06278-7, OCLC 246584055 (https://www.worldcat.org/oclc/246584055)
- Nick M (2005), *Al Jazari: The Ingenious 13th Century Muslim Mechanic* (http://www.alshindagah.com/marapr2005/jaziri.html), Al Shindagah, retrieved 16 October 2008.
- Nilsson N (30 October 2009). *The Quest for Artificial Intelligence*. Cambridge University Press. ISBN 978-0-52-112293-1.

- O'Connor KM (1994), *The alchemical creation of life (takwin) and other concepts of Genesis in medieval Islam* (http://repository.upenn.edu/dissertations/AAI9503804), University of Pennsylvania, pp. 1–435, retrieved 10 January 2007
- Olsen S (10 May 2004), *Newsmaker: Google's man behind the curtain* (http://news.cnet.com/Googles-man-behind-the-curtain/2008-1024_3-5208228.html), CNET, retrieved 17 October 2008.
- Olsen S (18 August 2006), *Spying an intelligent search engine* (http://news.cnet.com/Spying-an-intelligent-search-engine/2100-1032_3-6107048.html), CNET, retrieved 17 October 2008.
- Pearl J (1988), *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*, San Mateo, California: Morgan Kaufmann, ISBN 978-1-55860-479-7, OCLC 249625842 (https://www.worldcat.org/oclc/249625842).
- Poole D, Mackworth A, Goebel R (1998), *Computational Intelligence: A Logical Approach* (https://archive.org/details/computationalint00pool), Oxford University Press., ISBN 978-0-19-510270-3.
- Quevedo LT (1914), "Revista de la Academia de Ciencias Exacta", *Ensayos sobre Automática – Su definicion. Extension teórica de sus aplicaciones*, vol. 12, pp. 391–418
- Quevedo LT (1915), "Revue Génerale des Sciences Pures et Appliquées" (https://diccan.com/dicoport/Torres.htm), *Essais sur l'Automatique - Sa définition. Etendue théorique de ses applications*, vol. 2, pp. 601–611
- Randall B (1982), "From Analytical Engine to Electronic Digital Computer: The Contributions of Ludgate, Torres, and Bush" (http://www.fano.co.uk/ludgate/), *fano.co.uk*, retrieved 29 October 2018
- Russell SJ, Norvig P (2003), *Artificial Intelligence: A Modern Approach* (http://aima.cs.berkeley.edu/) (2nd ed.), Upper Saddle River, New Jersey: Prentice Hall, ISBN 0-13-790395-2.
- Russell SJ, Norvig P (2021). *Artificial Intelligence: A Modern Approach* (4th ed.). Hoboken: Pearson. ISBN 978-0-13-461099-3. LCCN 20190474 (https://lccn.loc.gov/20190474).
- Samuel AL (July 1959), "Some studies in machine learning using the game of checkers" (https://web.archive.org/web/20160303191010/http://domino.research.ibm.com/tchjr/journalindex.nsf/600cc5649e2871db852568150060213c/39a870213169f45685256bfa00683d74?OpenDocument), *IBM Journal of Research and Development*, **3** (3): 210–219, CiteSeerX 10.1.1.368.2254 (https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.368.2254), doi:10.1147/rd.33.0210 (https://doi.org/10.1147%2Frd.33.0210), S2CID 2126705 (https://api.semanticscholar.org/CorpusID:2126705), archived from the original (http://domino.research.ibm.com/tchjr/journalindex.nsf/600cc5649e2871db852568150060213c/39a870213169f45685256bfa00683d74?OpenDocument) on 3 March 2016, retrieved 20 August 2007.
- Saygin AP, Cicekli I, Akman V (2000), "Turing Test: 50 Years Later" (https://web.archive.org/web/20110409073501/http://crl.ucsd.edu/~saygin/papers/MMTT.pdf) (PDF), *Minds and Machines*, **10** (4): 463–518, doi:10.1023/A:1011288000451 (https://doi.org/10.1023%2FA%3A1011288000451), hdl:11693/24987 (https://hdl.handle.net/11693%2F24987), S2CID 990084 (https://api.semanticscholar.org/CorpusID:990084), archived from the original (http://crl.ucsd.edu/~saygin/papers/MMTT.pdf) (PDF) on 9 April 2011, retrieved 7 January 2004. Reprinted in Moor (2003, pp. 23–78).
- Searle J (1980), "Minds, Brains and Programs" (https://web.archive.org/web/20071210043312/http://members.aol.com/NeoNoetics/MindsBrainsPrograms.html), *Behavioral and Brain Sciences*, **3** (3): 417–457, doi:10.1017/S0140525X00005756 (https://doi.org/10.1017%2FS0140525X00005756), archived from the original (http://members.aol.com/NeoNoetics/MindsBrainsPrograms.html) on 10 December 2007, retrieved 13 May 2009.
- Simon HA, Newell A (1958), "Heuristic Problem Solving: The Next Advance in Operations Research", *Operations Research*, **6**: 1–10, doi:10.1287/opre.6.1.1 (https://doi.org/10.1287%2Fopre.6.1.1).
- Simon HA (1965), *The Shape of Automation for Men and Management*, New York: Harper & Row.
- Skillings J (2006), *Newsmaker: Getting machines to think like us* (http://news.cnet.com/Getting-machines-to-think-like-us---page-2/2008-11394_3-6090207-2.html?tag=st.next), CNET, retrieved 8 October 2008.
- Tascarella P (14 August 2006), "Robotics firms find fundraising struggle, with venture capital shy" (http://www.bizjournals.com/pittsburgh/stories/2006/08/14/focus3.html?b=1155528000%5E1329573), *Pittsburgh Business Times*, retrieved 15 March 2016.
- Turing A (1936–1937), "On Computable Numbers, with an Application to the Entscheidungsproblem" (http://www.abelard.org/turpap2/tp2-ie.asp), *Proceedings of the London Mathematical Society*, 2, **42** (42): 230–265, doi:10.1112/plms/s2-42.1.230 (https://doi.org/10.1112%2Fplms%2Fs2-42.1.230), S2CID 73712 (https://api.semanticscholar.org/CorpusID:73712), retrieved 8 October 2008.
- Turing A (October 1950), "Computing Machinery and Intelligence", *Mind*, **LIX** (236): 433–460, doi:10.1093/mind/LIX.236.433 (https://doi.org/10.1093%2Fmind%2FLIX.236.433), ISSN 0026-4423 (https://www.worldcat.org/issn/0026-4423).
- Turkle S (1984). *The second self: computers and the human spirit*. Simon and Schuster. ISBN 978-0-671-46848-4. OCLC 895659909 (https://www.worldcat.org/oclc/895659909).

- Wason PC, Shapiro D (1966). "Reasoning" (https://archive.org/details/newhorizonsinpsy0000foss). In Foss, B. M. (ed.). *New horizons in psychology*. Harmondsworth: Penguin. Retrieved 18 November 2019.
- Weizenbaum J (1976), *Computer Power and Human Reason*, W.H. Freeman & Company, ISBN 978-0-14-022535-8, OCLC 10952283 (https://www.worldcat.org/oclc/10952283).